

A sixth-order Lie-group Gauss collocation integrator for lower-pair mechanisms with position-, velocity- and acceleration-level constraints

Jingquan Wang^{a,*}

^a*University of Wisconsin–Madison, Department of Mechanical Engineering, 1513 University Avenue, Madison, WI, 53706, United States*

Abstract

Multibody systems with lower-pair joints in absolute coordinates are index-3 differential-algebraic equations on a Lie group. We study Gauss6/FullVA, a three-stage Gauss collocation step whose stage system enforces the joint constraints at position, velocity and acceleration level together with the Newton–Euler balance, and reconstructs the rotational endpoint through the exponential map. The central result is an exact algebraic identity: the lifted reduced Gauss stage, obtained by collocating the joint-coordinate equations of motion and lifting the result to absolute coordinates, satisfies every one of the 132 implemented stage rows, and conversely every root of the non-dynamic rows on the regular branch is such a lift. The method is therefore reduced Gauss collocation executed in absolute coordinates. Everything the implementation adds, the velocity-level endpoint closure and the inexact Newton solve, is an $O(h^7)$ perturbation, which gives a sixth-order error bound whose only standing hypothesis is smoothness on a compact neighbourhood of the trajectory; the solver and inverse interfaces are derived for $h \leq h_0$, and h_0 is quantified and traced to the curvature of the friction law. The row identities, the perturbation chain and the Butcher conditions are machine-checked in Lean 4 with Mathlib (57 theorems). Numerically the method is sixth order on a frictional two-body chain and on the ASME double pendulum, reproduces the driven ASME mechanisms to roundoff, is two to seven orders of magnitude more accurate than the one- and two-stage members of the same

*Corresponding author.

Email address: jingquanw@wisc.edu (Jingquan Wang)

family at the same Newton work, and reaches errors below 10^{-10} on the double pendulum where the public absolute-coordinate codes have errors of order one. A sweep of the Stribeck velocity locates the regime in which a sharp friction law reduces the observed order on coarse grids.

Keywords: Lie group integrator, index-3 DAE, multibody dynamics, lower-pair mechanisms, Gauss collocation, machine-checked proof

1. Introduction

Multibody systems with lower-pair joints are most conveniently modelled in absolute coordinates: each body carries its own position and rotation, and every joint contributes algebraic constraints. The price is an index-3 differential-algebraic system on a Lie group, and the integrator has to keep three things consistent at once: the constraints at position level, their time derivatives at velocity and acceleration level, and the rotation-group structure of the configuration. Methods that enforce only the position constraints let the velocity and acceleration constraints drift; methods that stabilize them by projection or index reduction change the equations that are being solved; and methods that integrate a reduced set of joint coordinates require the reduced equations of motion, which are mechanism-specific to derive and lose the direct access to reaction forces that the absolute formulation provides.

Time finite elements on Lie groups [1] have recently shown that high order is attainable for frictional index-3 pendulum problems, and the absolute-coordinate formulations of Negrut and co-workers [30, 31, 32, 33] have established a public benchmark suite of four mechanisms on which such methods can be compared. What has been missing is a method of order higher than four for the full index-3 problem in absolute coordinates whose order can be proved rather than observed, and whose proof survives the details that implementations add: acceleration-level constraint rows, Lagrange multipliers as stage unknowns, a Newton solver with a finite tolerance, and an endpoint reconstruction through the exponential map.

This paper studies such a method, Gauss6/FullVA. Its one-step map solves a single square nonlinear system in which the three-stage Gauss collocation conditions for the joint coordinates, the joint constraints at position, velocity and acceleration level, and the Newton–Euler balance at every stage appear as rows (Section 4); the rotational endpoint is then reconstructed from the Gauss quadrature of the body angular velocities through the exponential

map. The stage system has 132 rows for the two-body chain used throughout, and the same construction applies to the four benchmark mechanisms up to the joint list.

The central result is algebraic. We show that the lifted reduced Gauss stage, that is, the stage vector obtained by taking the three-stage Gauss collocation solution of the reduced joint-coordinate equations of motion and lifting it to absolute coordinates, makes every one of the 132 rows vanish, and conversely that any root of the non-dynamic rows on the regular branch is such a lift (Lemma 1). The implemented method is therefore reduced Gauss collocation, executed in absolute coordinates without ever forming the reduced equations, and the classical sixth-order theory of Gauss collocation applies to it verbatim. Everything that the implementation adds on top of the reduced collocation step, namely the endpoint closure of the velocity-level constraints and the inexact Newton solve, is a perturbation whose size we bound explicitly. The resulting local error bound (Theorem 1) is $C_{\text{loc}}h^7 = (C_G + C_E + C_N c_\eta)h^7$, and its only standing hypothesis is the smoothness of the constrained problem on a compact neighbourhood of the trajectory: the uniform invertibility of the stage Jacobian and the reachability of the $c_\eta h^7$ solver tolerance follow from smoothness for $h \leq h_0$ (Lemmas 3 and 7). We quantify h_0 on the benchmark and find that it is set by the curvature of the friction law, not by the collocation structure.

The algebraic identities and the perturbation lemmas are machine-checked in Lean 4 with Mathlib: the row identities are proved for a transcription of the implemented residual, the perturbation chain is proved with its explicit constants, and the Gauss tableau is shown to satisfy Butcher’s simplifying assumptions $B(6)$, $C(3)$, $D(3)$ and to fail $B(7)$. Butcher’s order theorem and the classical Gauss local-error estimate enter as cited results. The development ships with the paper (Appendix C).

Numerically, the method is sixth order on a frictional two-body chain over its regular branch for six step sizes (Section 6) and on the ASME double pendulum, the one benchmark mechanism of the public suite whose motion is not prescribed by a drive; the three driven mechanisms, including the two closed loops, are reproduced to roundoff (the driven pendulum from $h = 0.025$ on, the loops at every step size). The sixth-order stage system is repaid in work/precision terms against the lower-order members of the same Gauss family, which cost the same number of Newton iterations per step, and the double pendulum is compared on the public horizon with the public absolute-coordinate codes against one common reference. A sweep of

the friction regularization locates the regime in which a sharp Stribeck law reduces the observed order on coarse grids, and the constraint norms and the energy of the frictionless variant are followed over the whole regular branch. What we do not do is reproduce the temporal finite-element method of [1] on its own problems; the comparison with that method is at the level of formal order and is stated as such.

Contributions.

1. The exact stage identity: the FullVA stage system with acceleration-level constraints is reduced three-stage Gauss collocation lifted to absolute coordinates, in both directions.
2. A sixth-order local and global error theorem for the implemented one-step map under a smoothness assumption alone, with the solver and inverse interfaces derived and the small-step threshold quantified.
3. A Lean 4/Mathlib development of 57 theorems covering the row identities, the perturbation chain with explicit constants, and the tableau order conditions.
4. Convergence, work/precision, friction-regime and constraint-drift results on a frictional chain and on the four public benchmark mechanisms, all generated from a single implementation path and reproducible from the accompanying repository.

Section 2 reviews Lie-group and index-3 integrators for multibody systems. Section 3 fixes the setting and notation. Section 4 defines the one-step map. Section 5 contains the identity, the perturbation lemmas and the theorem. Section 6 reports the experiments, Section 7 the limitations, and Section 8 concludes.

2. Related work

Lie-group integrators. Integrators that advance rotations in a local tangent chart and return to the group through the exponential map go back to Crouch and Grossman [11] and to the Runge–Kutta–Munthe-Kaas construction [12, 13]; the surveys of Iserles et al. [14] and of Celledoni, Marthinsen and Owren [15] and the monograph of Hairer, Lubich and Wanner [5] give the general theory. In multibody dynamics the same viewpoint underlies the Lie-group generalized- α schemes of Brüls, Cardona and Arnold [19, 20], whose convergence for constrained systems was analysed in [18, 21], BDF schemes

on Lie groups [23], the geometrically exact beam and rigid-body formulations on $SE(3)$ [22], and singularity-free quaternion integration [24, 25]. Bottasso and Borri [17] discuss the interaction between the rotation parametrization and the integrator. These methods establish how to evolve rotations geometrically; they are of order at most two, or, in the Runge–Kutta–Munthe-Kaas case, are formulated for ordinary differential equations on the group rather than for the constrained index-3 problem.

Constrained mechanical systems and index-3 DAEs. Absolute-coordinate multibody systems are index-3 differential-algebraic equations [3, 7, 8]. Their numerical order depends not only on the integrator but on how the constraints and their differentiated forms are enforced: index reduction with stabilization [9], projection, and the review of constraint-enforcement approaches by Bauchau and Laulusa [10] describe the options and their failure modes. Runge–Kutta methods for index-2 and overdetermined index-3 formulations were analysed by Jay [27, 28, 29]; the Lobatto IIIA–IIIB pairs of that work are the closest relatives of the present stage system in that they enforce the position and velocity constraints at every stage. Lunk and Simeon [26] and Arnold and Brüls [18] treat the Newmark and generalized- α families for constrained systems. The absolute-coordinate formulations of Negrut and co-workers [30, 31, 32, 33, 34] compare three rotation parametrizations and a half-implicit scheme on a public suite of four mechanisms; that suite and its public code are used in Section 6. Reduced (joint-coordinate) formulations, for which the constrained problem becomes an ordinary differential equation, are classical [35, 36, 37]; the present paper shows that a particular absolute-coordinate collocation step coincides with Gauss collocation of such a reduced formulation without forming it.

Collocation and variational integrators. Gauss collocation is the standard high-order implicit Runge–Kutta family: the s -stage method has order $2s$, is symplectic and B -stable, and its stages are the values of a collocation polynomial [4, 6, 5]. Variational and symplectic integrators [38, 39, 40, 16] derive the discrete equations from a discrete action and treat holonomic constraints at the discrete level; Betsch and Steinmann [41] obtain energy–momentum conserving time finite elements for constrained systems. The present method is not variational: it enforces the constraint hierarchy pointwise at the stages and leaves the conservation properties to those of Gauss collocation on the reduced problem.

Friction and nonsmooth dynamics. Regularized friction laws such as the Brown–McPhee model [42] replace the set-valued Coulomb law by a smooth velocity-dependent force; Pennestrì et al. [43] review the alternatives. Non-smooth time-stepping and complementarity formulations [44, 45, 46] handle unilateral contact and Coulomb friction without regularization but are of low order by design. The problems in this paper are bilateral lower-pair mechanisms with regularized friction; the effect of the regularization parameter on the observed order is quantified in Section 6.

Time finite elements. Time finite elements for structural and rigid-body dynamics [47, 48, 49, 2, 50, 51] provide a different route to high order. The Lie-group time finite-element method of Chaturvedi, Sandu and Sandu [1] extends this route to frictional index-3 multibody DAEs; its Gauss–Lobatto family of degree m has formal order $2m - 1$, so that its $m = 3$ member is a fifth-order formula, although the source paper reports order reduction on the DAE problem. We do not reimplement that method; the comparison with it in this paper is at the level of formal order.

3. Problem setting

Equations of motion. A rigid body b has centre-of-mass position $r_b \in \mathbb{R}^3$, rotation $Q_b \in SO(3)$, translational velocity $v_b = \dot{r}_b$, body-frame angular velocity ω_b with $\dot{Q}_b = Q_b \widehat{\omega}_b$, mass m_b and body-frame inertia J_b . With $q = (r_b, Q_b)_b$, $v = (v_b, \omega_b)_b$ and $a = \dot{v}$, a mechanism with lower-pair joints is governed by

$$M\dot{v} + G(v) + \Phi_q(q)^T \lambda = F(q, v, t), \quad (1)$$

$$\dot{q} = \mathcal{K}(q)v, \quad (2)$$

$$\Phi(q) = 0, \quad (3)$$

where $M = \text{diag}(m_b I, J_b)$, G collects the gyroscopic terms $\omega_b \times J_b \omega_b$, Φ stacks the joint constraints, λ are the Lagrange multipliers, and $\mathcal{K}$ is the kinematic map, which is the identity on the translational part and $\omega_b \mapsto Q_b \widehat{\omega}_b$ on the rotational part. The system is an index-3 differential-algebraic equation on $(\mathbb{R}^3 \times SO(3))^N$. Differentiating the constraint once and twice gives the velocity- and acceleration-level constraints

$$\Phi_q(q)v = 0, \quad \Phi_q(q)a + \dot{\Phi}_q(q, v)v = 0, \quad (4)$$

which the exact solution satisfies identically; the method of Section 4 enforces Eq. (3) and both equations of Eq. (4) at every stage, which is what the acronym FullVA (full velocity- and acceleration-level enforcement) refers to.

Rotations. Rotations are represented by unit quaternions in the implementation and advanced by the retraction

$$Q_{n+1} = Q_n \text{Exp}(\eta_{n+1}), \quad Q_i = Q_n \text{Exp}(\eta_i), \quad (5)$$

with $\eta \in \mathbb{R}^3 \cong \mathfrak{so}(3)$ the local rotation increment and Exp the exponential map. The right-trivialized differential of the exponential map is written $J_r(\eta)$, so that $\dot{\eta} = J_r^{-1}(\eta) \omega$ along a motion with body angular velocity ω .

Lower pairs and joint coordinates. A lower pair j connects a parent body $p(j)$ (possibly the ground) and a child body $c(j)$. Let $P_j(q)$ and $C_j(q)$ be the attachment points on the two bodies, $d_j(q) = C_j(q) - P_j(q)$ the relative displacement, $a_j(q)$ the joint axis carried by the parent, $\tilde{a}_j(q)$ the axis carried by the child, and $\omega_j^{\text{rel}} = Q_{c(j)}\omega_{c(j)} - Q_{p(j)}\omega_{p(j)}$ the relative angular velocity in the world frame. Each pair carries a fixed world direction n_j , the joint axis at $t = 0$, and a fixed orthonormal pair $e_{j,1}, e_{j,2} \perp n_j$. A pair with two degrees of freedom (sliding along and rotating about its axis) has the joint coordinates

$$s_j(q) = n_j \cdot d_j(q), \quad \theta_j(q) = \text{atan2}(\tilde{t}_j \cdot (a_j \times t_j), \tilde{t}_j \cdot t_j), \quad (6)$$

where $t_j, \tilde{t}_j$ are twist reference vectors fixed in the parent and in the child. Their rates are $\dot{s}_j = n_j \cdot \dot{d}_j$ and $\dot{\theta}_j = a_j \cdot \omega_j^{\text{rel}}$, and their second rates are $\ddot{s}_j = n_j \cdot \ddot{d}_j$ and $\ddot{\theta}_j = a_j \cdot \dot{\omega}_j^{\text{rel}} + \dot{a}_j \cdot \omega_j^{\text{rel}}$. Revolute, prismatic and spherical joints are obtained by fixing one or both coordinates or by dropping the alignment rows; the constraint rows of the four benchmark mechanisms are listed in Appendix A.

The two-body chain. The main benchmark is a two-body open chain (Table 1). Body 0 is connected to the ground by a pair with sliding and rotation along a fixed ground axis; body 1 is connected to body 0 by a second pair whose axis $a_1(q) = Q_0 \hat{a}_0^{\text{next}}$ is fixed in body 0. The two axes fixed in body 0 are not parallel and body 0 spins, so $a_1(q)$ moves on a cone about the ground axis while n_1 and $e_{1,m}$ stay frozen at their $t = 0$ values. The second pair is therefore a four-constraint lower pair with a fixed sliding direction and a moving rotation axis, not a cylindrical pair in the strict sense; its

friction sliding speed is $\dot{s}_1(n_1 \cdot a_1)$, and the modelling is regular as long as $n_1 \cdot a_1(q) \neq 0$. On the initial state of Table 1 this factor decreases from 1 to 0.34 on $[0, 0.5]$ and vanishes at $t \approx 0.605$, where the frozen sliding direction ceases to define the pair; all chain experiments are run on the regular branch $[0, 0.5]$. Each joint carries gravity, constant external loads, viscous damping and a regularized Brown–McPhee friction torque and force [42] with static and dynamic coefficients μ_s, μ_d and Stribeck velocity v_s ; the friction force enters the Newton–Euler rows through the multipliers (normal force) and the sliding speed, and is C^∞ in the state for $v_s > 0$. The smooth case uses $v_s = 0.5$; Section 6 varies v_s down to 0.02.

Table 1: Two-body chain benchmark. Directions are given before normalization; the initial rotations are the unique rotations mapping the body-fixed axis and twist vectors of each pair onto their parent-side counterparts.

Quantity	Value
Masses m_0, m_1 [kg]	2.7, 1.9
Inertias J_0, J_1 [kg m ²]	diag(0.11, 0.24, 0.31), diag(0.08, 0.19, 0.27)
Ground axis, ground twist	(0.31, 0.78, 0.54), first vector of an orthonormal complement
Parent-side axes $\hat{a}_0^{\text{prev}}, \hat{a}_1^{\text{prev}}$	(0.18, 0.91, -0.37), (-0.24, 0.87, 0.43)
Child-side axes $\hat{a}_0^{\text{next}}, \hat{a}_1^{\text{next}}$	(0.42, 0.69, 0.59), (0, 1, 0)
Attachment points $s_0^{\text{prev}}, s_1^{\text{prev}}$ [m]	(0.12, -0.04, 0.18), (-0.08, 0.05, 0.16)
Attachment points $s_0^{\text{next}}, s_1^{\text{next}}$ [m]	(0.16, 0.02, -0.20), (0, 0, 0)
Gravity [m/s ²]	(0, 0, -9.81)
External forces (world) [N]	(0.8, -0.35, 0.25), (-0.25, 0.45, -0.10)
External torques (body) [N m]	(0.03, -0.02, 0.015), (-0.018, 0.026, -0.014)
Friction μ_s, μ_d , viscous coefficient, radius	0.30, 0.20, 0.030, 1.0
Stribeck velocity v_s	0.5 (smooth case); 0.2, 0.1, 0.05, 0.02 in the sweep
Initial $s_j(0), \dot{s}_j(0), \dot{\theta}_j(0)$	(0.35, 0.42), (0.42, -0.26), (0.18, -0.14)

Notation. Throughout, h is the step size, $t_n = nh$, $x_n = (q_n, v_n)$ the endpoint state (accelerations and multipliers are stage quantities), c_i, b_i, A_{ik} ($i, k = 1, 2, 3$) the nodes, weights and matrix of the three-stage Gauss tableau, Z the stacked stage vector, and $\|\cdot\|$ a fixed norm on the finite-dimensional

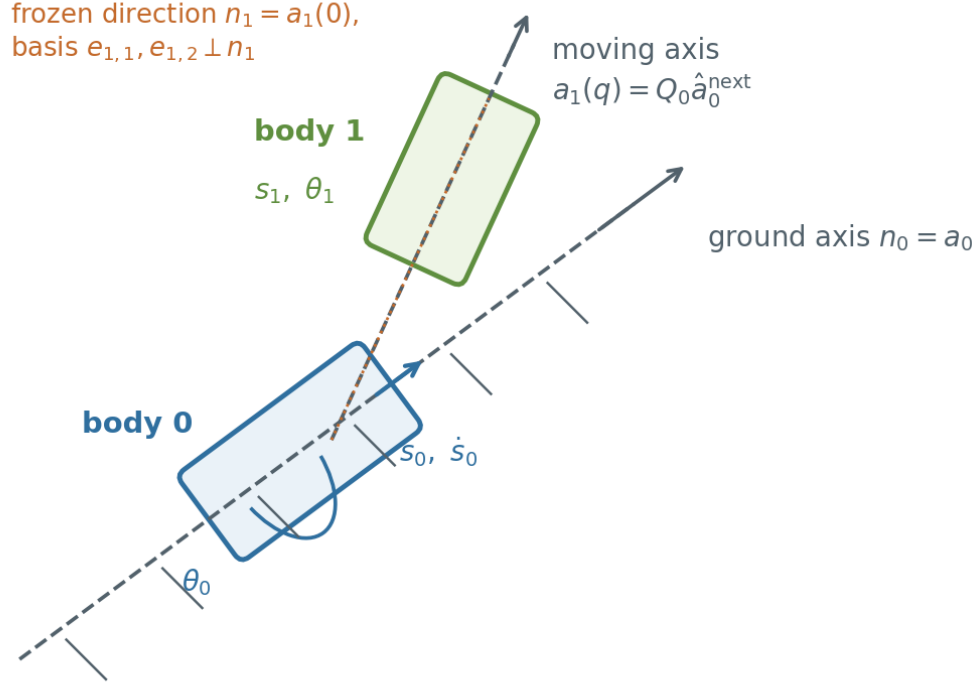


Figure 1: The two-body chain: body 0 slides and spins along the fixed ground axis, body 1 slides and spins along the axis $a_1(q)$ carried by body 0; the second pair keeps its sliding direction n_1 and its basis $e_{1,m}$ frozen at their $t = 0$ values.

space at hand. Constants C with subscripts are independent of h and of the number of steps.

4. The Gauss6/FullVA step

4.1. Stage unknowns and stage system

For each stage $i = 1, 2, 3$ and each body b the unknowns are the translational position $r_{b,i}$, the rotation increment $\eta_{b,i}$ with $Q_{b,i} = Q_{b,n} \text{Exp}(\eta_{b,i})$, the

velocities $v_{b,i}, \omega_{b,i}$ and the accelerations $a_{b,i}, \alpha_{b,i}$; each lower pair j contributes its multipliers $\lambda_{j,i}$, four per pair for the chain. With two bodies and two pairs the stage vector

$$Z = (Z_1, Z_2, Z_3), \quad Z_i = (r_i, \eta_i, v_i, \omega_i, a_i, \alpha_i, \lambda_i), \quad (7)$$

has $3(2 \cdot 18 + 2 \cdot 4) = 132$ components.

Write $\psi_j(q) = \tilde{a}_j(q) - a_j(q)$ for the axis-alignment residual and, for a joint-coordinate function g with rate $\dot{g}$,

$$\Delta_i[g] = g(Z_i) - g(x_n) - h \sum_{k=1}^3 A_{ik} \dot{g}(Z_k) \quad (8)$$

for its Gauss collocation defect at stage i . With $t_i = t_n + c_i h$, the applied wrench $(f_{b,i}, \tau_{b,i})$ evaluated at the stage, and the total force and torque $F_{b,i}^r = f_{b,i} + \Phi_{r,b}(q_i)^T \lambda_i$, $T_{b,i}^\theta = \tau_{b,i} + \Phi_{\theta,b}(q_i)^T \lambda_i$ including the multiplier wrench, the stage system reads, for every pair j , every $m = 1, 2$ and every body b ,

$$\begin{aligned} 0 &= e_{j,m} \cdot d_j(q_i), & 0 &= e_{j,m} \cdot \psi_j(q_i), \\ 0 &= e_{j,m} \cdot \dot{d}_j(q_i, v_i), & 0 &= e_{j,m} \cdot \dot{\psi}_j(q_i, v_i), \\ 0 &= e_{j,m} \cdot \ddot{d}_j(q_i, v_i, a_i), & 0 &= e_{j,m} \cdot \ddot{\psi}_j(q_i, v_i, a_i), \\ 0 &= \Delta_i[s_j], & 0 &= \Delta_i[\dot{s}_j], \\ 0 &= \Delta_i[\theta_j], & 0 &= \Delta_i[\dot{\theta}_j], \\ 0 &= m_b a_{b,i} - F_{b,i}^r, & 0 &= J_b \alpha_{b,i} + \omega_{b,i} \times J_b \omega_{b,i} - T_{b,i}^\theta. \end{aligned} \quad (9)$$

The first three lines are the lower-pair constraints at position, velocity and acceleration level (12 rows per pair and stage); the fourth and fifth lines are Gauss collocation of the four joint coordinates $(s_j, \dot{s}_j, \theta_j, \dot{\theta}_j)$ (4 rows per pair and stage); the last line is the Newton–Euler balance (6 rows per body and stage). Table 2 gives the inventory: 44 rows per stage for the 44 unknowns, $72 + 24 + 36 = 132$ over the three stages. The absolute body coordinates $(r_i, \eta_i, v_i, \omega_i)$ carry no collocation row of their own; they are determined by the constraint rows once the joint coordinates are collocated. The implementation projects the two translational collocation rows onto the current joint axis $a_j(q_i)$ rather than onto n_j ; on the constraint manifold $d_j, \dot{d}_j, \ddot{d}_j \parallel n_j$, so the two forms differ by the factor $n_j \cdot a_j(q_i)$, nonzero on the regular branch, and define the same stage solution. Driven mechanisms add prescribed right-hand sides to the three constraint levels of the driven row. The multiplier wrench collects the joint normal forces, the axis-alignment torques and the friction force along the joint axis; the friction law enters only there.

Table 2: Row inventory of the stage system for the two-body chain (per stage and in total).

Row family	Per pair or body	Per stage	Total
Position-level constraints $\Phi = 0$	4 per pair	8	24
Velocity-level constraints $\Phi_q v = 0$	4 per pair	8	24
Acceleration-level constraints $\Phi_q a + \ddot{\Phi}_q v = 0$	4 per pair	8	24
Joint-coordinate collocation $\Delta_i[s_j], \Delta_i[\dot{s}_j], \Delta_i[\theta_j], \Delta_i[\dot{\theta}_j]$	4 per pair	8	24
Newton–Euler balance	6 per body	12	36
Unknowns $(r, \eta, v, \omega, a, \alpha)$ per body, λ per pair	18, 4	44	132

4.2. Endpoint reconstruction and closure

After the stage solve the endpoint is reconstructed with the Gauss weights,

$$r_{n+1} = r_n + h \sum_{i=1}^3 b_i v_i, \quad v_{n+1} = v_n + h \sum_{i=1}^3 b_i a_i, \quad (10)$$

$$Q_{n+1} = Q_n \text{Exp} \left(h \sum_{i=1}^3 b_i J_r^{-1}(\eta_i) \omega_i \right), \quad \omega_{n+1} = \omega_n + h \sum_{i=1}^3 b_i \alpha_i. \quad (11)$$

The reconstructed endpoint satisfies the position-level constraints only up to the quadrature error ($O(h^7)$ per step, Lemma 4) and the velocity-level constraints likewise. The endpoint closure map $\mathcal{C}_h$ then projects the endpoint velocities (v_{n+1}, ω_{n+1}) onto the velocity-level constraint manifold of Eq. (4) at q_{n+1} by the minimal-norm correction, that is, by solving the KKT system with matrix $\begin{bmatrix} I & \Phi_q^T \\ \Phi_q & 0 \end{bmatrix}$; positions and rotations are not projected. The correction is an $O(h^7)$ perturbation of the step (Lemma 5). The closed endpoint $x_{n+1} = \mathcal{C}_h(\mathcal{E}_h(Z))$, with $\mathcal{E}_h$ the map of Eqs. (10)–(11), defines the one-step map $\Psi_h(x_n) = x_{n+1}$.

4.3. Newton solve and predictor

The stage system $F_h(Z; x_n) = 0$ is solved by Newton’s method,

$$D_Z F_h(Z^{(k)}; x_n) \Delta Z = -F_h(Z^{(k)}; x_n), \quad Z^{(k+1)} = Z^{(k)} + \Delta Z, \quad (12)$$

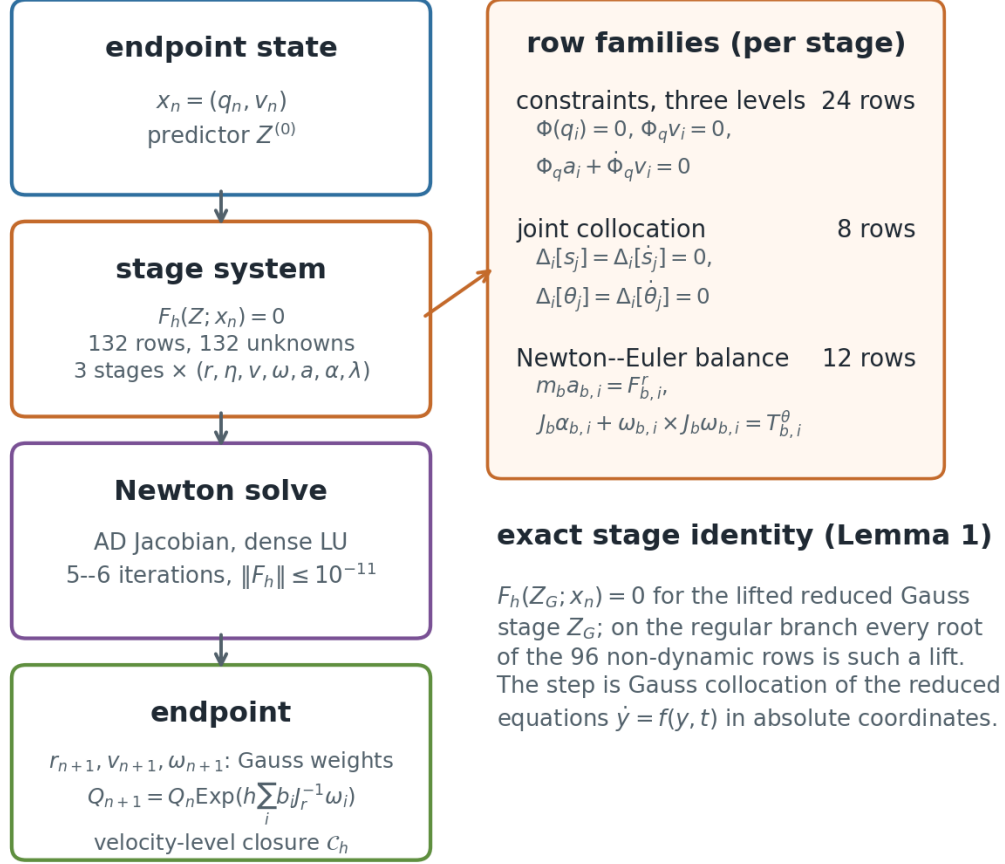


Figure 2: One step of Gauss6/FullVA: the stage system couples the three constraint levels, the joint-coordinate collocation rows and the Newton–Euler balance; the endpoint is reconstructed with the Gauss weights and the exponential map and closed at velocity level.

with the 132×132 Jacobian assembled by forward-mode automatic differentiation of the single residual function and factorized densely. The iteration stops when the residual norm or the update norm falls below the tolerance $\tau_N = 10^{-11}$ and is declared failed after $k_{\max} = 80$ iterations. The predictor extrapolates positions and velocities from the endpoint state with the constant endpoint acceleration, sets the rotation increments, angular velocities and angular accelerations of the stages to zero, and estimates the multipliers from the endpoint force balance; the consequences of this choice for the convergence analysis are discussed after Lemma 7. On the chain the solve takes

between five and six Newton iterations per step at every step size (Section 6).

Table 3: One step of Gauss6/FullVA.

Step	Operation
1	From $x_n = (q_n, v_n)$ and h , form the predictor $Z^{(0)}$.
2	Assemble the stage residual $F_h(Z; x_n)$ of Eq. (9): constraint rows at the three levels, joint-coordinate collocation rows, Newton–Euler rows.
3	Iterate Eq. (12) with the automatic-differentiation Jacobian until $\ F_h\ \leq \tau_N$ or $\ \Delta Z\ \leq \tau_N$.
4	Reconstruct the endpoint by Eqs. (10)–(11).
5	Apply the velocity-level closure $\mathcal{C}_h$; record the Newton count and the endpoint constraint norms.

Cost. One Newton iteration costs one residual and Jacobian evaluation and one dense 132×132 factorization; a step costs five to six such iterations. The Jacobian is block-sparse (each stage couples to the others only through the collocation rows), which the present reference implementation does not exploit. The two- and one-stage members of the same family, obtained by replacing the Gauss tableau, have 88 and 44 unknowns and are used for the work/precision comparison of Section 6.

5. Error analysis

The analysis has three parts. Lemma 1 shows that the stage system of Eq. (9) is exactly three-stage Gauss collocation of the reduced joint-coordinate equations of motion, written in absolute coordinates; the stage residual vanishes identically at the lifted reduced Gauss stage and no stage-residual perturbation term appears. The two perturbations the implementation does introduce, the endpoint closure and the inexact Newton solve, are bounded by $O(h^7)$ (Lemmas 5 and 6), and the local defect is transferred to the grid by a discrete Gronwall argument (Lemma 8). The hypotheses are collected in Assumption 1; Lemmas 3 and 7 show that the second and third of them follow from the first for small steps. The algebraic identities and the perturbation lemmas are machine-checked in Lean 4/Mathlib (Appendix C); Butcher’s order theorem for the Gauss tableau and the classical Gauss local-error estimate [3] are used as cited results.

5.1. Reduced chart, lift and hypotheses

Let $y = (s_0, \theta_0, s_1, \theta_1, \dot{s}_0, \dot{\theta}_0, \dot{s}_1, \dot{\theta}_1)$ collect the joint coordinates of Eq. (6) and their rates. On the smooth constrained branch the equations of motion reduce to

$$\dot{y} = f(y, t), \quad (13)$$

where f returns the rates and the joint accelerations obtained from the constrained Newton–Euler equations. Let φ_h denote the flow of Eq. (13) over one step, $\mathcal{L}(y, t) = (q, v, a, \lambda)$ the smooth lift to absolute coordinates given by forward kinematics together with the velocities, accelerations and multipliers of the constrained dynamics, and Y_1, Y_2, Y_3 the Gauss collocation stages of Eq. (13) from y_n ,

$$Y_i = y_n + h \sum_{k=1}^3 A_{ik} f(Y_k, t_k), \quad i = 1, 2, 3. \quad (14)$$

The lifted reduced Gauss stage is $Z_G = (\mathcal{L}(Y_1), \mathcal{L}(Y_2), \mathcal{L}(Y_3))$. We write $F_h(\cdot; x_n)$ for the implemented 132-row residual, $\mathcal{E}_h$ for the endpoint reconstruction of Eqs. (10)–(11), $\mathcal{C}_h$ for the endpoint closure, $\mathcal{P}_h = \mathcal{C}_h \circ \mathcal{E}_h$, and $\Psi_h^G = \mathcal{E}_h(Z_G)$ for the reconstructed Gauss endpoint. Norms are fixed finite-dimensional norms on the reduced chart, the stage space and the endpoint space; chart-to-configuration norm equivalences on the compact tube are absorbed into the constants.

Assumption 1 (Regularity and perturbation scale). *Fix a final time T and a compact reduced-state tube K containing the exact trajectory of Eq. (13) at distance at least $d_K > 0$ from ∂K . The following hold uniformly for $y \in K$, $0 \leq t \leq T$ and $0 < h \leq h_0$, with constants independent of h and of the number of steps.*

- (H1) (Smooth branch.) *The constraint Jacobian Φ_q has constant full row rank, the constrained mass matrix is nonsingular on the tangent space, f is C^7 , the lift $\mathcal{L}$ is smooth on the tube, and the joint coordinates complete the constraints to a chart, i.e. the map $q \mapsto (\Phi(q), s_0(q), \theta_0(q), s_1(q), \theta_1(q))$ has an invertible Jacobian.*
- (H2) (Local inverses and stability.) *The stage Jacobian $J_h = D_Z F_h(Z_G; x_n)$ is invertible with $\|J_h^{-1}\| \leq M$ and $\|D_Z^2 F_h\| \leq L$ on a ball $B_\rho(Z_G)$ with $ML\rho \leq \frac{1}{2}$. The closure subsystem E of Lemma 5 is C^2 on a ball $B_{r_E}(z_*)$ around a same-branch closed anchor z_* with $E(z_*) = 0$; $DE(z_*)$ has a*

right inverse B_* with $\|B_*\| \leq M_E$, $\|DE(z) - DE(z_*)\| \leq L_E\|z - z_*\|$ on that ball, $M_E L_E r_E \leq \frac{1}{2}$, the unclosed endpoint $z_u = \Psi_h^G$ lies in $B_{r_E/2}(z_*)$, and its raw closure defect satisfies $\|E(z_u)\| \leq C_{E,\text{raw}} h^7$. The one-step map satisfies

$$\|\Psi_h(y) - \Psi_h(\bar{y})\| \leq (1 + C_s h)\|y - \bar{y}\|, \quad y, \bar{y} \in K. \quad (15)$$

(H3) (Solver envelope.) The computed Newton iterate $\tilde{Z}$ lies in the ball $B_\rho(Z_G)$ of (H2) and its residual satisfies $\|F_h(\tilde{Z}; x_n)\| \leq \eta_h \leq c_\eta h^7$. The output map satisfies $\|D_Z \mathcal{P}_h\| \leq M_N$ on that ball.

(H1) is the compact-tube regularity hypothesis of classical collocation theory. (H2) and (H3) are the interfaces through which the implementation enters; Lemmas 3 and 7 derive them from (H1) for $h \leq h_0$, and they are kept as separate items only because the theorem is stated for any mechanism whose stage system has the form of Eq. (9). (In the accompanying repository the three hypotheses are called P1, P2 and P6.) The theorem also uses the classical Gauss local-error estimate: for $f \in C^7$ on the compact tube there is $C_{G,0}$ with

$$\|u(t_n + h) - \varphi_h(y_n)\| \leq C_{G,0} h^7, \quad (16)$$

where u is the degree-three collocation polynomial through Eq. (14) [3]. The tableau used in the implementation satisfies Butcher's simplifying assumptions $B(6)$, $C(3)$, $D(3)$ and fails $B(7)$, so its order is exactly six; these algebraic conditions are machine-checked for the coefficients hard-coded in the implementation.

5.2. Exact stage identity

Lemma 1 (Exact stage identity). *Under (H1), the implemented 132-row residual satisfies*

$$F_h(Z_G; x_n) = 0. \quad (17)$$

Conversely, on the branch where $n_j \cdot a_j(q_i) \neq 0$, every stage vector that satisfies the 96 non-dynamic rows of Eq. (9) lies on the constraint manifold with consistent velocities and accelerations and has joint coordinates satisfying Eq. (14). Hence the stage solve is three-stage Gauss collocation of Eq. (13) in the joint-coordinate chart, written in absolute coordinates.

Proof. The 72 constraint rows vanish on Z_G because each $\mathcal{L}(Y_k)$ lies on the constraint manifold with consistent velocities and accelerations. The 24 joint-coordinate rows are, by Eq. (6), the reduced collocation equations (14) read

component-wise, because $s_j, \theta_j, \dot{s}_j, \dot{\theta}_j$ are coordinates of y and $\ddot{s}_j, \ddot{\theta}_j$ are components of f ; the implemented projection onto $a_j(q_i)$ multiplies each translational row by $n_j \cdot a_j(q_i)$. The 36 Newton–Euler rows vanish because $\mathcal{L}$ supplies the stage accelerations and multipliers satisfying force and moment balance, and the implemented rows are exactly these balance defects with the implemented wrench sign conventions. For the converse, the constraint rows force $d_j, \dot{d}_j, \ddot{d}_j \parallel n_j$, so each translational collocation row factors as $(s_j(Z_i) - s_j(x_n) - h \sum_k A_{ik} \dot{s}_j(Z_k)) (n_j \cdot a_j(q_i))$, and the twist rows are already scalar. Both directions are verified row by row for a transcription of the implemented residual in the Lean development. $\square$

The lemma replaces any stage-residual perturbation estimate: the residual at the lifted Gauss stage is not small, it is zero. What remains to be controlled is which root Newton selects, how the endpoint is reconstructed and closed, and how inexact the Newton solve is. The lemma is stated for the open chain, where the joint coordinates form a global chart; for closed loops the same argument applies on any local chart of independent coordinates. The rows of Eq. (9) are transcribed from the single implemented residual function, whose automatic derivative is the Newton Jacobian, so there is one residual and Jacobian path; the Lean development transcribes the same rows, and the binding between transcription and source is a static check, not a theorem. Numerically, the converged stage vectors of the chain runs read in the joint-coordinate chart have reduced collocation defects and out-of-axis components of $d_j, \dot{d}_j, \ddot{d}_j$ at roundoff (Section 6).

Lemma 2 (Branch selection). *Under (H2), Z_G is the unique zero of $F_h(\cdot; x_n)$ in $B_\rho(Z_G)$. In particular the exact stage root on that branch equals Z_G , and $\mathcal{E}_h(Z_G) = \Psi_h^G$.*

Proof. For $Z \in B_\rho(Z_G)$ the mean value inequality gives $\|F_h(Z) - F_h(Z_G) - J_h(Z - Z_G)\| \leq \frac{L\rho}{2} \|Z - Z_G\|$. If $F_h(Z) = 0$, then Eq. (17) gives $\|Z - Z_G\| \leq M \|F_h(Z) - F_h(Z_G) - J_h(Z - Z_G)\| \leq \frac{ML\rho}{2} \|Z - Z_G\| \leq \frac{1}{4} \|Z - Z_G\|$, so $Z = Z_G$. $\square$

Lemma 3 ((H2) from (H1) for small steps). *Assume (H1). Then there is $h_0 > 0$ such that for $0 < h \leq h_0$: (i) the stage Jacobian $J_h = D_Z F_h(Z_G)$ is invertible with*

$$\|J_h^{-1}\| \leq 2\|J_0^{-1}\|, \quad J_0 = \lim_{h \rightarrow 0} J_h, \quad \|J_h - J_0\| \leq C_J h, \quad (18)$$

(ii) the closure Jacobian is invertible, so B_* exists with $\|B_*\|$ bounded uniformly, (iii) the raw closure defect satisfies $\|E(z_u)\| \leq C_{E,\text{raw}}h^7$, and (iv) the one-step map satisfies Eq. (15). Hence all of (H2) follows from (H1) for small steps.

Proof. At $h = 0$ the collocation rows reduce to $s_j(Z_i) - s_j(x_n)$, $\theta_j(Z_i) - \theta_j(x_n)$ and their rate analogues, so J_0 is block diagonal over the stages and block triangular within a stage: the position block is the Jacobian of $q \mapsto (\Phi, s, \theta)$, the velocity block is the same matrix acting on (v, ω) because on the constraint manifold the rate rows are the time derivatives of the coordinate rows, and the acceleration block is the constrained Newton–Euler system $\begin{bmatrix} M & \Phi_q^T \\ \Phi_q & 0 \end{bmatrix}$ acting on (a, α, λ) . Each block is invertible under (H1), so J_0 is invertible with an h -independent inverse bound. The only h -dependence of F_h is the factor $h \sum_k A_{ik}$ in the collocation rows, so $\|J_h - J_0\| \leq C_J h$ on the compact tube, and for $2\|J_0^{-1}\|C_J h \leq 1$ Eq. (18) follows from the Neumann series. The endpoint closure is a square KKT system in the endpoint velocities whose matrix is $\begin{bmatrix} I & \Phi_q^T \\ \Phi_q & 0 \end{bmatrix}$, invertible by full row rank of Φ_q ; its raw defect is the velocity-level constraint value at the Gauss-weight endpoint, which is the quadrature error of Lemma 4 and hence $O(h^7)$; the implicit function theorem gives a same-branch closed anchor z_* within $O(h^7)$ of z_u , so $z_u \in B_{r_E/2}(z_*)$ for small h . Finally, by Lemma 1 the one-step map read in the reduced chart is the reduced Gauss step, whose derivative is $I + O(h)$ by the implicit function theorem for the stage equations, composed with the smooth endpoint reconstruction and an $O(h^7)$ closure correction; this gives the Lipschitz scale $1 + C_s h$. $\square$

The threshold h_0 of the Neumann argument is not sharp. On the chain (Table B.10 in Appendix B) the sufficient condition $\|J_0^{-1}\| \|J_h - J_0\| \leq \frac{1}{2}$ would require $h \leq 2 \times 10^{-5}$ in the Euclidean norm on the stage layout and $h \leq 8 \times 10^{-4}$ in the endpoint-linearized norm $\|J_0^{-1} \cdot\|$, whereas the conclusion $\|J_h^{-1}\| \leq 2\|J_0^{-1}\|$ holds on every grid used with ratio at most 1.55. The gap comes from the curvature of the friction law: the largest entries of $(J_h - J_0)/h$ sit in the Newton–Euler rows against the stage velocities, where the second derivative of the Brown–McPhee law with $v_s = 0.5$ enters, and the same mechanism without joint friction has C_J smaller by a factor of about fifteen and $h_0 \approx 1.4 \times 10^{-2}$. The friction law, not the collocation structure, sets the small-step threshold; Section 6 shows the same dependence in the observed orders when v_s is decreased.

5.3. Endpoint defect, closure and inexact Newton

Lemma 4 (Gauss endpoint defect). *Under (H1) there is a constant C_G , uniform on the compact tube, such that*

$$\|\Psi_h^G(y) - \varphi_h(y)\| \leq C_G h^7. \quad (19)$$

Proof. Let u be the collocation polynomial of Eq. (14) and g any component of the lifted state along u , so that $g(t) = \rho(u(t))$ for a smooth chart function ρ and $g'(t_k)$ equals the lifted stage velocity at Y_k ; for the rotational component, g is the Lie-algebra increment $\eta(t)$ with $\eta' = J_r^{-1}(\eta)\omega$ along the lifted trajectory. The reconstruction Eqs. (10)–(11) replaces $g(t_n + h) - g(t_n) = \int_{t_n}^{t_n+h} g'(t) dt$ by the Gauss quadrature $h \sum_i b_i g'(t_i)$. Because the Gauss weights integrate polynomials of degree at most five exactly, a Peano-kernel argument gives

$$\left| \int_{t_n}^{t_n+h} g'(t) dt - h \sum_{i=1}^3 b_i g'(t_i) \right| \leq \frac{h^7}{60} \max_{[t_n, t_n+h]} |g^{(7)}|, \quad (20)$$

with $g^{(7)}$ bounded uniformly on the tube by (H1) and the smoothness of u . Hence each reconstructed endpoint component equals the lifted endpoint of u up to $O(h^7)$, and Exp is smooth. Combining with Eq. (16) and the chart norm equivalence gives Eq. (19). The bound Eq. (20) is machine-checked. $\square$

Lemma 5 (Endpoint-closure perturbation). *Let $E(z) = 0$ be the velocity-level KKT closure subsystem solved by $\mathcal{C}_h$, and let $z_u = \Psi_h^G$ be the unclosed endpoint. Under (H2) the right-inverse Newton iteration $z_{l+1} = z_l - B_* E(z_l)$, $z_0 = z_u$, converges to a closed endpoint $z_E = \mathcal{C}_h(z_u)$ with $E(z_E) = 0$, and any closed endpoint of the form $z_u + B_* w$ in $B_{r_E}(z_*)$ satisfies*

$$\|z_E - z_u\| \leq 2M_E \|E(z_u)\| \leq C_E h^7, \quad C_E = 2M_E C_{E, \text{raw}}. \quad (21)$$

Proof. Write $G(w) = E(z_u + B_* w)$. Since $DE(z_*)B_* = I$ and $\|DE(z) - DE(z_*)\| \leq L_E \|z - z_*\| \leq L_E r_E$ on the ball, the mean value inequality gives $\|G(w_1) - G(w_2) - (w_1 - w_2)\| \leq M_E L_E r_E \|w_1 - w_2\| \leq \frac{1}{2} \|w_1 - w_2\|$ whenever both arguments stay in the ball. The map $w \mapsto w - G(w)$ is therefore a $\frac{1}{2}$ -contraction of the ball of radius $2\|E(z_u)\|$ into itself, which stays inside $B_{r_E}(z_*)$ for $h \leq h_0$; its fixed point w_* gives $z_E = z_u + B_* w_*$ with $E(z_E) = 0$. For any root of this form, the same inequality with $w_2 = 0$ yields $\|w\| \leq 2\|E(z_u)\|$, hence $\|z_E - z_u\| = \|B_* w\| \leq 2M_E \|E(z_u)\|$. The raw-defect bound of (H2) completes Eq. (21). $\square$

Lemma 6 (Inexact Newton). *Under (H2) and (H3),*

$$\begin{aligned} \|\tilde{Z} - Z_G\| &\leq 2M\eta_h, \\ \|\mathcal{P}_h(\tilde{Z}) - \mathcal{P}_h(Z_G)\| &\leq C_N\eta_h \leq C_N c_\eta h^7, \quad C_N = 2MM_N. \end{aligned} \tag{22}$$

Proof. With Eq. (17) and the mean value inequality of Lemma 2, $\|\tilde{Z} - Z_G\| \leq M(\|F_h(\tilde{Z})\| + \frac{L\rho}{2}\|\tilde{Z} - Z_G\|)$, so $\|\tilde{Z} - Z_G\| \leq 2M\eta_h$. The output bound follows from $\|D_Z\mathcal{P}_h\| \leq M_N$ on the ball and the envelope $\eta_h \leq c_\eta h^7$. $\square$

Lemma 7 (Newton reaches the solver envelope). *Under (H1) and (H2) there is $h_0 > 0$ such that for $0 < h \leq h_0$ the lifted endpoint predictor $Z_*^{(0)} = (\mathcal{L}(x_n), \mathcal{L}(x_n), \mathcal{L}(x_n))$ lies in $B_{\rho/2}(Z_G)$; for every predictor $Z^{(0)} \in B_{\rho/2}(Z_G)$ the simplified Newton iteration $Z^{(k+1)} = Z^{(k)} - J_h^{-1}F_h(Z^{(k)})$ stays in $B_\rho(Z_G)$ and*

$$\|Z^{(k)} - Z_G\| \leq 2^{-k}\|Z^{(0)} - Z_G\|, \quad \|F_h(Z^{(k)})\| \leq L_F 2^{-k}\|Z^{(0)} - Z_G\|, \tag{23}$$

where L_F is a Lipschitz constant of F_h on the ball; the same decay holds for the full Newton iteration of Eq. (12). Consequently the stopping rule $\|F_h(Z^{(k)})\| \leq c_\eta h^7$ is met after $k = O(\log(1/h))$ iterations, and (H3) holds by construction whenever the stage solve starts in $B_{\rho/2}(Z_G)$.

Proof. The stages of Z_G are $\mathcal{L}(Y_k)$ with $\|Y_k - y_n\| = O(h)$ on the compact tube, so $\|Z_*^{(0)} - Z_G\| = O(h)$ and $Z_*^{(0)} \in B_{\rho/2}(Z_G)$ for small h . The linearization inequality of (H2) makes the simplified Newton map a $\frac{1}{2}$ -contraction of $B_\rho(Z_G)$ into itself with fixed point Z_G (Lemma 2), which gives the first bound in Eq. (23); the second follows from $F_h(Z_G) = 0$ and the Lipschitz bound. For the full Newton iteration, (H2) gives $\|D_Z F_h(Z) - J_h\| \leq L\|Z - Z_G\| \leq L\rho$ on the ball, so the perturbation bound of Lemma 3 yields $\|D_Z F_h(Z)^{-1}\| \leq 2M$, and the standard estimate $\|Z^{(k+1)} - Z_G\| \leq ML\|Z^{(k)} - Z_G\|^2 \leq \frac{1}{2}\|Z^{(k)} - Z_G\|$ gives the same decay. The simplified-Newton decay estimate is machine-checked. $\square$

The predictor of the implementation (Section 4) is not $Z_*^{(0)}$: it sets the angular-velocity and angular-acceleration guesses to zero, so its distance to Z_G is of the order of $\|\omega_n\| + \|\alpha_n\|$ rather than $O(h)$ (about 50 on the chain, Table B.11), and the first full Newton step from it expands the distance by 12 to 18% before the quadratic phase sets in. The lemma therefore covers the implemented run through the ball membership of (H3), which is confirmed

a posteriori on every grid used: the converged roots are the Gauss roots to roundoff. Replacing the zero angular guesses by the endpoint values would make the implemented predictor an $O(h)$ predictor and bring the run under the lemma directly.

The experiments of Section 6 use the fixed tolerance $\tau_N = 10^{-11}$ rather than the h^7 -scaled rule. On the chain sweep the residual of the accepted iterate is between 10^{-14} and 8×10^{-12} at every step and step size (Table B.12); relative to h^7 this is $c_\eta \leq 10^{-3}$ for $h \geq 0.05$ but $c_\eta \approx 10^6$ at $h = 0.003125$, so the fixed-tolerance runs satisfy (H3) only with a large constant on the finest grids, and the inexact-Newton term $C_N c_\eta h^7$ of the theorem is then a pessimistic bound: the observed errors keep decreasing as h^6 down to 10^{-13} , three orders of magnitude below $2M\|F_h(\tilde{Z})\|$. One additional Newton iteration per step would bring the accepted residual to roundoff at every step size; the h^7 -scaled rule with $c_\eta = 10^4$ has been exercised on the chain and gives the same observed orders as the fixed tolerance.

5.4. Local-to-global transfer and main theorem

Lemma 8 (Local-to-global transfer). *Let Ψ_h be a one-step map on the reduced chart, $t_n = nh$, and $y_{n+1} = \Psi_h(y_n)$ with y_0 the exact initial state. Suppose that for $0 \leq n < N$ with $Nh \leq T$ the exact states $Y_n = \varphi_{t_n}(y_0)$ satisfy $\|\Psi_h(Y_n) - Y_{n+1}\| \leq C_\ell h^{p+1}$, that $\|\Psi_h(y_n) - \Psi_h(Y_n)\| \leq (1 + C_s h)\|y_n - Y_n\|$ whenever $y_n \in K$, and that $C_\ell \Gamma_s(T) h^p \leq d_K$, where $\Gamma_s(T) = (e^{C_s T} - 1)/C_s$ for $C_s > 0$ and $\Gamma_s(T) = T$ for $C_s = 0$. Then $y_n \in K$ for all $n \leq N$ and*

$$\max_{0 \leq n \leq N} \|y_n - Y_n\| \leq C_\ell \Gamma_s(T) h^p. \quad (24)$$

If moreover the reporting map $\mathcal{R}$ (chart to reported position and velocity) is $C_{\mathcal{R}}$ -Lipschitz on K , then $\max_n \|\mathcal{R}(y_n) - \mathcal{R}(Y_n)\| \leq C_{\mathcal{R}} C_\ell \Gamma_s(T) h^p$.

Proof. Let $e_n = \|y_n - Y_n\|$. By induction, as long as $y_n \in K$, $e_{n+1} \leq (1 + C_s h)e_n + C_\ell h^{p+1}$, hence $e_n \leq C_\ell h^{p+1} \sum_{k < n} (1 + C_s h)^k$. For $C_s > 0$ the geometric sum equals $((1 + C_s h)^n - 1)/(C_s h) \leq (e^{C_s n h} - 1)/(C_s h) \leq \Gamma_s(T)/h$, and for $C_s = 0$ it equals $n \leq T/h$. Thus $e_n \leq C_\ell \Gamma_s(T) h^p \leq d_K$, which keeps y_n in the tube and closes the induction. The reporting bound is the Lipschitz property. The lemma is machine-checked. $\square$

Definition 1 (Branch-selected one-step map). *The one-step map Ψ_h is defined on the set $\mathcal{D}_h \subset K$ of reduced states for which the transition satisfies (H2) and (H3): the computed iterate lies in $B_\rho(Z_G)$ with residual at most*

η_h , the endpoint is reconstructed by Eqs. (10)–(11) and closed by $\mathcal{C}_h$, and $\Psi_h(y) = \mathcal{P}_h(\tilde{Z})$.

Theorem 1 (Sixth-order error bound). *Let Assumption 1 hold and let $C_{\text{loc}} = C_G + C_E + C_N c_\eta$ with C_G , C_E , C_N from Lemmas 4, 5 and 6. Then there is $h_0 > 0$ such that for $0 < h \leq h_0$ and every $y \in \mathcal{D}_h$,*

$$\|\Psi_h(y) - \varphi_h(y)\| \leq C_{\text{loc}} h^7. \quad (25)$$

Consequently, for every grid $t_n = nh$, $0 \leq n \leq N_h$, $N_h h \leq T$, whose transitions lie in $\mathcal{D}_h$ and whose numerical trajectory $y_{n+1} = \Psi_h(y_n)$ starts from the exact reduced initial state,

$$\max_{0 \leq n \leq N_h} \|y_n - y(t_n)\| \leq C_{\text{red}} h^6, \quad C_{\text{red}} = C_{\text{loc}} \Gamma_s(T), \quad (26)$$

and the reported position-velocity map $\mathcal{R}(y) = (q(y), v(y))$ satisfies

$$\max_{0 \leq n \leq N_h} \|\mathcal{R}(y_n) - \mathcal{R}(y(t_n))\| \leq C_{qv} h^6, \quad C_{qv} = C_{\mathcal{R}} C_{\text{red}}. \quad (27)$$

The constants depend on the tube, on T and on the constants of Assumption 1, including c_η , but not on h , N_h or the linear-algebra backend. Under (H1) alone the same conclusions hold for $h \leq h_0$ whenever the stage solve is started in $B_{\rho/2}(Z_G)$ and stopped by the rule $\|F_h\| \leq c_\eta h^7$ (Lemmas 3 and 7).

Proof. Fix $y \in \mathcal{D}_h$ and write $\Psi_h^G = \mathcal{E}_h(Z_G)$, $\Psi_h^E = \mathcal{P}_h(Z_G)$ and $\Psi_h(y) = \mathcal{P}_h(\tilde{Z})$. By Lemma 2 the exact stage root on the branch is Z_G , so there is no stage-root perturbation term, and

$$\begin{aligned} \|\Psi_h(y) - \varphi_h(y)\| &\leq \|\Psi_h^G(y) - \varphi_h(y)\| + \|\Psi_h^E(y) - \Psi_h^G(y)\| \\ &\quad + \|\Psi_h(y) - \Psi_h^E(y)\| \leq C_G h^7 + C_E h^7 + C_N c_\eta h^7 \end{aligned} \quad (28)$$

by Lemmas 4, 5 and 6, after shrinking h_0 so that all three apply on the tube. This is Eq. (25). Choose h_0 additionally so that $C_{\text{loc}} \Gamma_s(T) h_0^6 \leq d_K$. Lemma 8 with $p = 6$, $C_\ell = C_{\text{loc}}$ and the stability scale Eq. (15) gives Eq. (26); the reporting-map clause gives Eq. (27). The last sentence follows from Lemmas 3 and 7, which supply (H2) and (H3) from (H1) for small steps. $\square$

Two remarks on the scope of the theorem. First, it is an estimate for the reduced state and for the reported positions and velocities; it says nothing about the accuracy of the stage multipliers or reaction forces beyond what

follows from the smooth dependence of the lift on the reduced state, and it does not turn a small constraint residual on a closed-loop mechanism into a trajectory-error bound, which would need a stability constant for that mechanism. Second, because the stage residual vanishes identically at the lifted Gauss stage, the order of the method is that of reduced Gauss collocation; the absolute-coordinate formulation contributes the exact three-level constraint enforcement, the multipliers and reaction forces at every stage, the friction law through the multipliers, and the Lie-group endpoint reconstruction, but no additional stage-level approximation. The same argument gives order $2s$ for the s -stage Gauss member with the same stage system, which Section 6 confirms for $s = 1, 2$.

Formal-order comparison with time finite elements. The Lie-group time finite-element family of [1] with Gauss–Lobatto polynomials of degree m has formal order $2m - 1$ [2]; its $m = 3$ member is a fifth-order formula. The comparison of the present method with that family is at this formal level only: order six against order five for a stage system of comparable size. We have not implemented the time finite-element method and make no statement about its accuracy or cost on its own test problems.

6. Numerical experiments

All experiments use the implementation of Section 4 with the fixed Newton tolerance $\tau_N = 10^{-11}$ and the dense automatic-differentiation Jacobian; the scripts that produce every number and figure of this section are listed in Appendix D. Errors are measured against a reference solution computed with the same method at a step size h_{ref} much smaller than the finest step of the sweep; the difference between the references at h_{ref} and $2h_{\text{ref}}$ is reported as the resolution floor of the reference, and a fitted order uses only the step sizes whose errors exceed that floor by a factor of 100. Position and velocity errors are Frobenius norms over the bodies at the final time; the orientation error is the largest geodesic angle between the numerical and reference rotations, and the angular-velocity error the Frobenius norm of the body-frame difference.

6.1. Convergence on the frictional chain

The chain of Table 1 with $v_s = 0.5$ is integrated over its regular branch $[0, 0.5]$ with $h = 0.1/2^k$, $k = 0, \dots, 5$, against the reference at $h_{\text{ref}} = 0.1/256$

(1280 steps). Table 4 and Figure 3 show the result. All four error measures decrease by 5.5 to 6 orders of magnitude over the sweep; the fitted orders over the five step pairs above the floor are 6.25 (position), 6.28 (orientation), 6.11 (velocity) and 6.27 (angular velocity), and the consecutive pairwise orders settle at 5.98 and 5.99 on the two finest pairs. The endpoint velocity-level constraint norm is at roundoff on every step because of the closure; the position-level norm, which is not projected, decreases as h^6 from 10^{-7} to 3×10^{-15} as the theory predicts. The Newton solve takes 6.0 iterations per step on the coarse grids and 5.7 on the fine ones.

Table 4: Convergence on the frictional chain, $T = 0.5$, $v_s = 0.5$: errors at $t = T$ against the $h_{\text{ref}} = 0.1/256$ reference, pairwise order of the position error to the next finer step, and maximal endpoint constraint norms at position (Φ) and velocity ($\Phi_q v$) level; the number of steps is $0.5/h$. Resolution floor of the reference: 3×10^{-15} (position), 3×10^{-15} (orientation), 2×10^{-14} (velocity), 1×10^{-14} (angular velocity).

h	Newton/step	position	orientation	velocity	ang. velocity	order	$\ \Phi\ $	$\ \Phi_q v\ $
0.1	6.00	2.77e-4	8.77e-5	4.96e-4	1.10e-3	6.95	9.8e-8	7.7e-16
0.05	6.00	2.24e-6	7.98e-7	2.18e-6	9.47e-6	4.58	1.5e-9	1.3e-15
0.025	5.90	9.38e-8	2.82e-8	1.95e-7	3.59e-7	7.64	2.4e-11	1.2e-15
0.0125	5.72	4.69e-10	1.43e-10	9.42e-10	1.82e-9	5.98	3.7e-13	9.8e-14
0.00625	5.72	7.45e-12	2.28e-12	1.49e-11	2.89e-11	5.99	6.3e-15	9.6e-14
0.003125	5.72	1.17e-13	3.55e-14	2.33e-13	4.44e-13	–	3.0e-15	8.2e-15
fitted order		6.25	6.28	6.11	6.27			

Diagnostics of the analysis. The constants and hypotheses of Section 5 are measured on the same runs and reported in Appendix B: the converged stage vectors satisfy the exact stage identity to roundoff (Table B.9); the inverse bound of Lemma 3 holds with ratio at most 1.55 although its sufficient Neumann condition would need $h \leq 8 \times 10^{-4}$ (Table B.10); the implemented predictor sits at distance about 50 from the Gauss stage (Table B.11); and under the fixed tolerance the accepted residual is 10^{-14} to 8×10^{-12} , so that (H3) holds with c_η growing from 10^{-4} to 10^6 across the sweep (Table B.12).

6.2. The Gauss family: orders two, four and six at the same Newton work

Replacing the three-stage tableau by the two-stage Gauss tableau (order four) or by the implicit midpoint rule (order two) leaves the stage system of Eq. (9) unchanged except for its size (88 and 44 unknowns). Figure 4 and Table 5 compare the three members on the chain over $[0, 0.5]$ against the same reference. The fitted position orders are 2.10, 4.17 and 6.25, in agreement

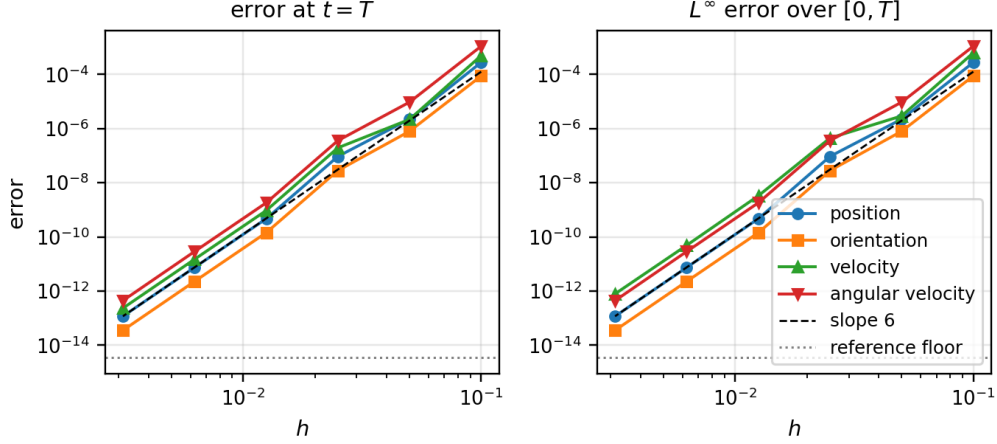


Figure 3: Convergence on the frictional chain over the regular branch $[0, 0.5]$: errors at $t = T$ (left) and over all grid points (right) against the $h_{\text{ref}} = 0.1/256$ reference, with the slope-6 line anchored at the finest position error and the resolution floor of the reference.

with the remark after Theorem 1. The Newton iteration count per step is the same for the three members (5.7 to 6.0), so at a given step size the work differs only by the size of the dense factorization; in the present Python implementation the wall time per step differs by at most a third across the family, being dominated by interpreter overhead. At $h = 0.0125$ the three-stage member is 1.6×10^5 times more accurate than the one-stage and 310 times more accurate than the two-stage member, and at $h = 0.003125$ the factors are 4×10^7 and 5×10^3 ; to reach the accuracy of the three-stage member at $h = 0.05$ (2×10^{-6}) the one-stage member needs $h \approx 2 \times 10^{-3}$, that is, about 23 times more steps and Newton iterations.

6.3. Sharpness of the friction law

The Brown–McPhee law is smooth for every $v_s > 0$, but its derivatives grow like v_s^{-k} , and Lemma 3 predicts a small-step threshold that shrinks accordingly. Figure 5 and Table 6 repeat the convergence sweep of Section 6.1 for $v_s \in \{0.5, 0.2, 0.1, 0.05, 0.02\}$, each against its own reference at $h_{\text{ref}} = 0.1/256$. Two things happen as v_s decreases. The fitted order over the whole sweep drops from 6.25 to 2.47, because the coarse grids are in a pre-asymptotic regime in which the error is dominated by the friction transition; and the step size at which the pairwise order returns to six moves from

Table 5: Gauss family with the same stage system on the chain, $T = 0.5$: position and velocity errors at $t = T$ and total Newton iterations. The wall time of the first row of each member includes just-in-time compilation and is omitted.

h	1 stage (order 2)			2 stages (order 4)		
	position	velocity	Newton	position	velocity	Newton
0.1	1.16e-2	6.53e-2	31	1.48e-3	3.23e-3	30
0.05	1.16e-3	1.21e-2	60	6.23e-5	8.38e-5	60
0.025	2.95e-4	2.97e-3	115	1.61e-6	2.81e-6	117
0.0125	7.41e-5	7.39e-4	229	1.45e-7	2.66e-7	229
0.00625	1.87e-5	1.85e-4	457	9.09e-9	1.67e-8	458
0.003125	4.68e-6	4.61e-5	915	5.69e-10	1.04e-9	915
0.0015625	1.17e-6	1.15e-5	1829	3.55e-11	6.51e-11	1830
0.00078125	2.93e-7	2.88e-6	3658	—	—	—
fitted order	2.10	2.04		4.17	4.17	

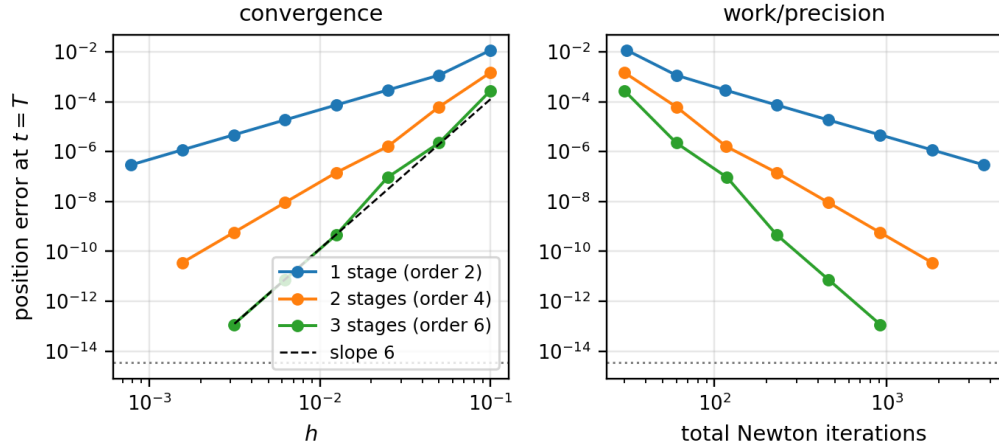


Figure 4: Gauss family with the same FullVA stage system on the chain, $T = 0.5$: position error against step size (left) and against total Newton iterations (right). Three stages (order six) are the method of this paper.

$h = 0.1$ to $h \approx 3 \times 10^{-3}$, where all five cases show pairwise orders between 5.3 and 6.7 on the finest pair. The measured Neumann threshold h_0 decreases from 7.6×10^{-4} to 7.1×10^{-6} over the same range, so the analysis predicts the trend correctly but its sufficient condition is two to three orders of magnitude more conservative than the observed recovery step. Every run in the sweep converged, with 5.7 to 6.4 Newton iterations per step.

Table 6: Friction sweep on the chain, $T = 0.5$: fitted position and velocity orders over the six step sizes, pairwise position orders from $h = 0.1$ to $h = 0.003125$, coarsest step at which the pairwise order reaches 5.5, and the Neumann threshold $h_0 = 1/(2C_J)$ in the endpoint-linearized norm measured over the first ten steps at $h = 0.01$.

v_s	position order	velocity order	pairwise position orders	recovery h	h_0 (L)
0.5	6.25	6.11	6.95, 4.58, 7.64, 5.98, 5.99	0.1	7.6e-4
0.2	4.84	4.74	2.11, 4.54, 2.74, 8.57, 5.85	0.0125	1.4e-4
0.1	4.74	4.78	0.47, 5.71, 4.14, 5.88, 6.72	0.05	6.7e-5
0.05	3.66	3.66	0.17, 4.11, 4.47, 3.46, 5.27	–	2.1e-5
0.02	2.47	2.48	-0.22, 2.29, 2.90, 1.55, 6.20	0.00625	7.1e-6

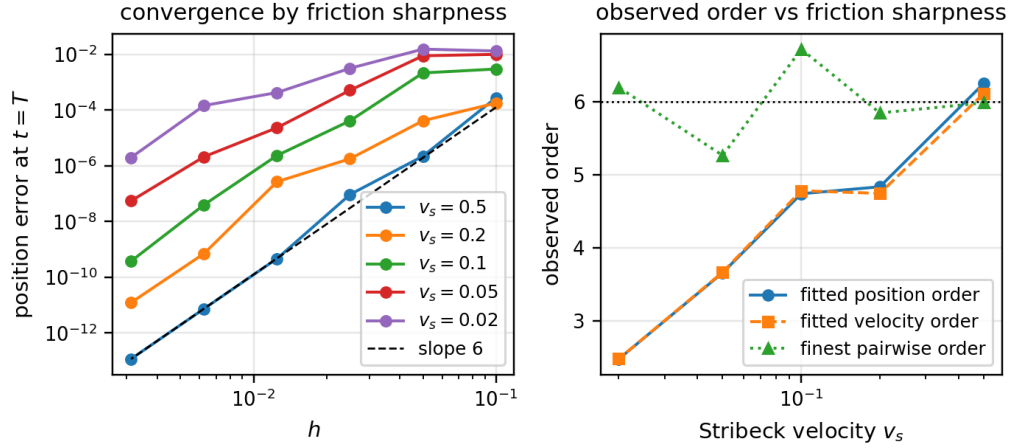


Figure 5: Friction sweep on the chain: position error against step size for five Stribeck velocities (left) and fitted orders against v_s (right).

6.4. The four benchmark mechanisms

The public benchmark suite of [32] consists of a driven single pendulum, a free double pendulum released from the horizontal, and two driven closed loops, a four-link mechanism and a slider-crank (Appendix A). Three of the four are kinematically driven: their motion is fixed by the drive constraint and the dynamics only determines the reactions. Gauss6/FullVA is run on all four over $T = 1$ with $h = 0.1/2^k$ (Figure 7, Table 7). For the driven single pendulum, whose exact motion is known in closed form, the final-state position errors are 5.9×10^{-10} , 9.3×10^{-12} and 1.5×10^{-13} at $h = 0.1$, 0.05 and 0.025 (pairwise orders 6.0 and 6.0) and at roundoff, 10^{-14} to 10^{-15} , for every finer step; its velocity errors scatter between 10^{-15} and 4×10^{-9} .

without trend in h , at the level set by the Newton tolerance of that harness, which does not project the endpoint velocities. For the two closed loops the trajectory errors against the $h = 0.1/128$ reference are between 6×10^{-16} and 1.3×10^{-14} at every step size from 0.05 down to 0.0016: the driven motion is reproduced to roundoff, the loop-closure constraints included, and no order can be measured. The double pendulum is the only genuine dynamics test of the suite. Its final-state errors against the $h = 0.1/128$ reference are 4.6×10^{-5} , 1.6×10^{-7} , 1.2×10^{-9} , 1.9×10^{-11} and 3.1×10^{-13} for $h = 0.1$ to 0.00625, with pairwise orders 8.2, 7.0, 6.0, 6.0 in position and 9.7, 7.0, 6.0, 5.9 in velocity; the coarsest pair is pre-asymptotic, the rest is the sixth order of Theorem 1. At $h = 0.003125$ the error, 2.3×10^{-14} , has reached the resolution floor of the reference (2×10^{-14} , the difference between the $h = 0.1/128$ and $h = 0.1/64$ runs).

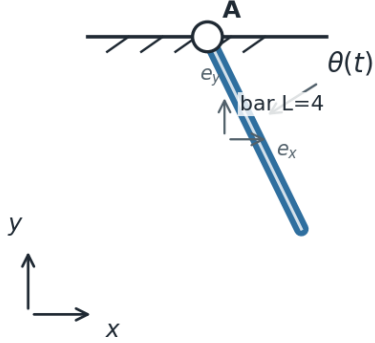
Table 7: The four benchmark mechanisms over $T = 1$: final-state position and velocity errors (single pendulum: against the analytic driven motion; double pendulum: against the $h = 0.1/128$ reference) and trajectory L^∞ errors of the closed loops against their $h = 0.1/128$ reference. The pairwise order refers to the position error and the next finer step; the double-pendulum reference has a resolution floor of 2×10^{-14} .

h	single pendulum (driven, analytic reference)			double pendulum (free)		
	position	velocity	order	position	velocity	order
0.1	5.94e-10	2.43e-9	6.0	4.59e-5	1.56e-4	8.2
0.05	9.32e-12	3.83e-11	6.0	1.58e-7	1.85e-7	7.0
0.025	1.45e-13	9.49e-11	–	1.24e-9	1.48e-9	6.0
0.0125	1.08e-14	6.77e-10	–	1.93e-11	2.29e-11	6.0
0.00625	1.58e-14	3.73e-9	–	3.10e-13	3.86e-13	3.8
0.003125	1.33e-15	1.05e-15	–	2.29e-14	6.15e-14	–
	four-link (driven loop)			slider-crank (driven loop)		
	position	velocity		position	velocity	
0.05–0.0016	0.9–1.3e-14	1.5–4.0e-14	–	0.6–1.2e-15	1.4–2.8e-15	–

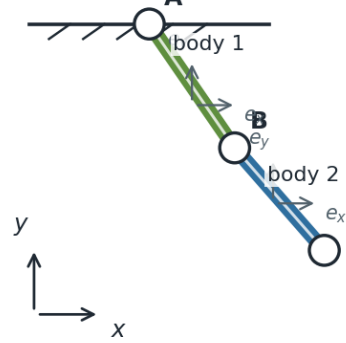
6.5. Comparison with the public absolute-coordinate codes

The public codes of the benchmark suite (the 2021 rA, rp and rε formulations of [32] and the 2022 half-implicit rA and rA-half codes of [33]) are run as published, in their dynamics mode with their own tolerance policies, on the double pendulum over the public horizon $T = 3$. Every error in Table 8 is the final-state L^∞ difference to one common reference, the local Gauss6/FullVA run at $h = 10^{-4}$ (30 000 steps). That the public model and the local one are the same mechanism is checked on the short horizon $T = 0.1$, where

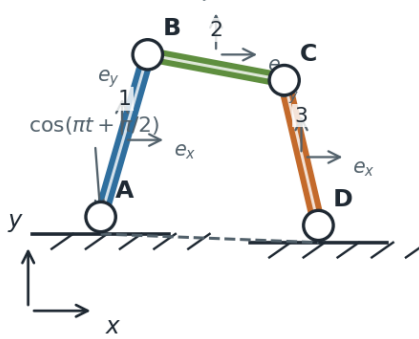
single pendulum
driven revolute joint, $\theta(t)$ prescribed



double pendulum
two revolute joints, released from rest



four-link loop
driven closed loop, revolute/universal/spherical joints



slider-crank

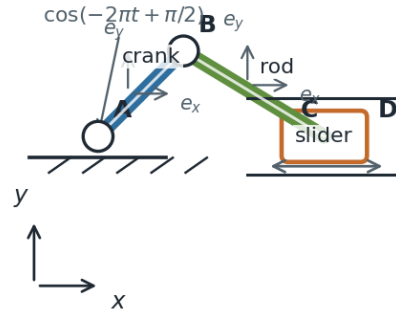


Figure 6: The four benchmark mechanisms of the public suite [32]; joint data in Appendix A.

the public rA final state differs from the local one by 3.7×10^{-3} at $h = 0.1/16$ and 9.2×10^{-4} at $h = 0.1/64$, a first-order decrease and not a constant offset.

Over $T = 3$ the double pendulum amplifies perturbations strongly, and the public codes do not resolve it at any of the step sizes: the rA, $r\epsilon$ and 2022 rA codes give identical results with position errors from 1.8 at $h = 0.1$ to 0.17 at $h = 0.001$ (pairwise orders 0.2 to 0.9, approaching first order only below $h = 0.002$); the rp code fails to converge for $h \geq 0.025$ and is at first order below; the half-implicit rA-half code does not converge at all on this

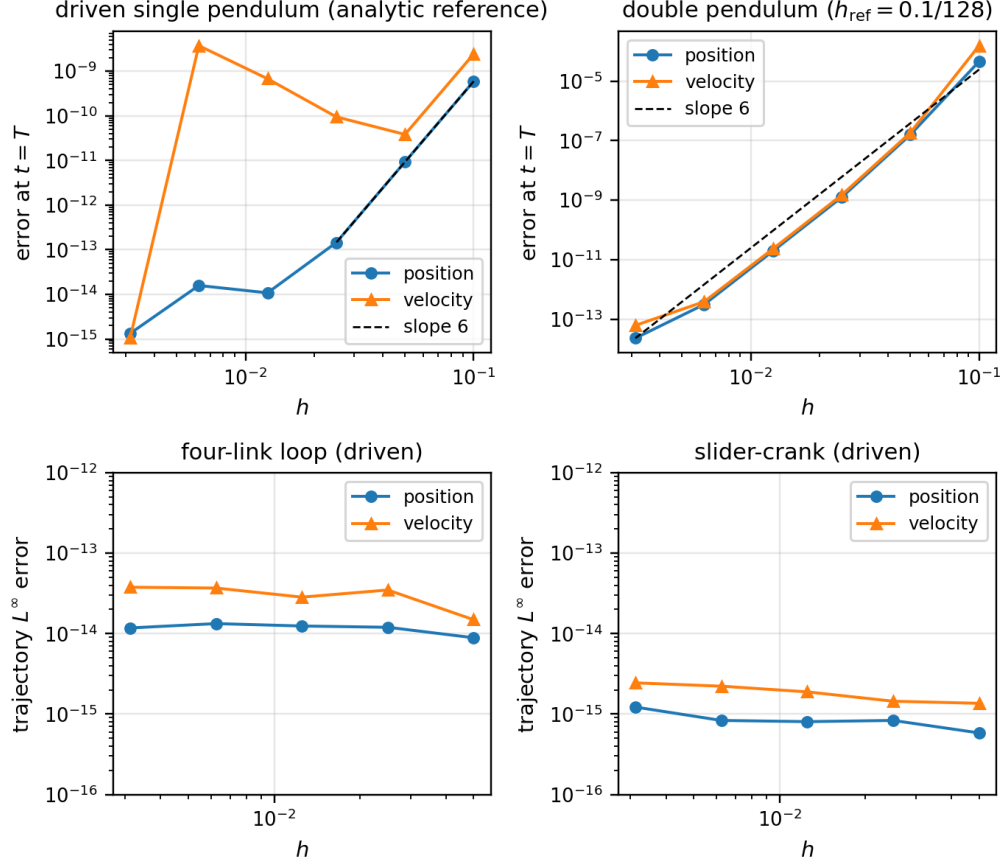


Figure 7: Gauss6/FullVA on the four benchmark mechanisms over $T = 1$. The three driven mechanisms are reproduced at roundoff for every step size; the double pendulum shows the sixth order.

horizon (errors between 0.9 and 5.8). Gauss6/FullVA reaches 4.3×10^{-4} at $h = 0.1$, 2.1×10^{-11} at $h = 0.01$ and the resolution floor of the reference, 2×10^{-13} , at $h \leq 0.002$, with pairwise orders 7.2, 9.2, 6.0 and 6.1 on the first four pairs. The Newton counts are 3.0 to 4.5 iterations per step for Gauss6/FullVA against 1.8 to 6.8 for the 2021 codes and 3 to 32 for the 2022 codes; wall times are not comparable across the implementations (the local harness is a Python reference implementation with a dense automatic-differentiation Jacobian) and are omitted.

On the three driven mechanisms the public horizon adds nothing to Sec-

tion 6.4: Gauss6/FullVA reproduces the driven motion to 10^{-14} over $T = 3$ at $h = 10^{-2}$, 10^{-3} and 10^{-4} , while the public rA code, measured against its own kinematic-mode reference, has position errors below 10^{-9} but velocity errors that decrease at first order, 1.1×10^{-1} , 1.1×10^{-2} , 1.1×10^{-3} on the four-link mechanism and 8.5×10^{-3} , 8.7×10^{-4} , 8.8×10^{-5} on the slider-crank.

Table 8: Double pendulum over the public horizon $T = 3$: final-state L^∞ errors against the common Gauss6/FullVA reference at $h = 10^{-4}$. The 2021 rε and the 2022 half-implicit rA codes return the same numbers as the 2021 rA code. “fail”: the public Newton iteration did not converge. The reference has a resolution floor of about 2×10^{-13} .

h	Gauss6/FullVA		2021 public rA		2021 public rp	2022 rA-half
	position	velocity	position	velocity	position	position
0.1	4.29e−4	1.51e−3	1.84	6.04	fail	5.76
0.05	3.03e−6	9.84e−6	1.58	5.03	fail	5.01
0.025	5.34e−9	1.72e−8	1.34	4.13	fail	1.34
0.0125	8.12e−11	2.66e−10	1.06	3.43	0.749	0.903
0.01	2.10e−11	7.09e−11	0.958	3.40	1.25	1.12
0.002	2.56e−13	1.76e−12	0.319	1.10	0.233	4.26
0.001	2.16e−13	1.66e−12	0.169	0.532	0.113	1.92

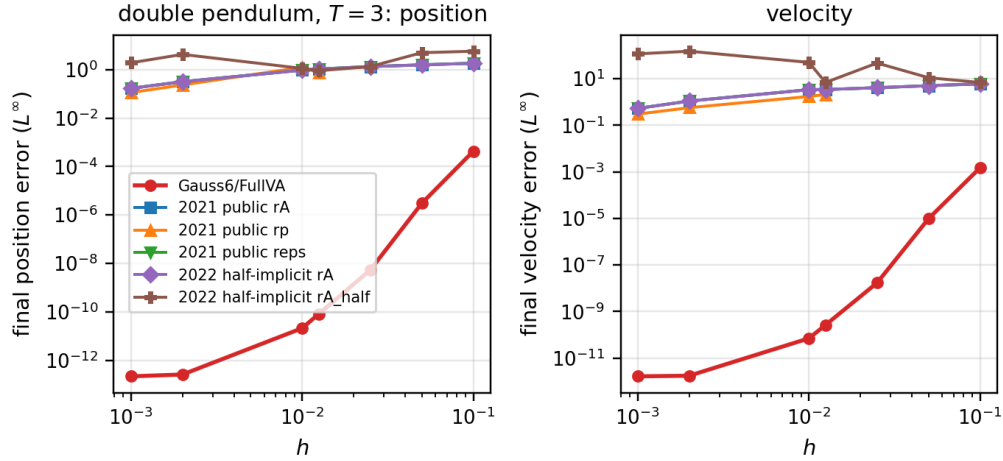


Figure 8: Double pendulum over the public horizon $T = 3$: final position (left) and velocity (right) errors of Gauss6/FullVA and of the public absolute-coordinate codes against the common $h = 10^{-4}$ reference.

6.6. Constraint drift and energy over the regular branch

Figure 9 follows the chain over the whole regular branch $[0, 0.5]$ at $h = 0.01$ in two variants: the frictional benchmark, and a conservative variant with $\mu_s = \mu_d = 0$, no viscous damping and no external loads, for which the total mechanical energy is an invariant of the exact flow. The endpoint position- and velocity-level constraint norms stay at 1.1×10^{-13} and 9.5×10^{-14} in both variants, the acceleration-level norms at the stages below 2×10^{-14} , and the quaternion unit error at 2×10^{-16} ; the relative energy error of the conservative variant is 4.3×10^{-12} at every step (the exact Gauss step is symplectic on the reduced problem, and the endpoint closure perturbs it by $O(h^7)$), while the frictional variant dissipates 37% of its initial energy over the branch. The Newton count is 5 per step without friction and 5.7 with it.

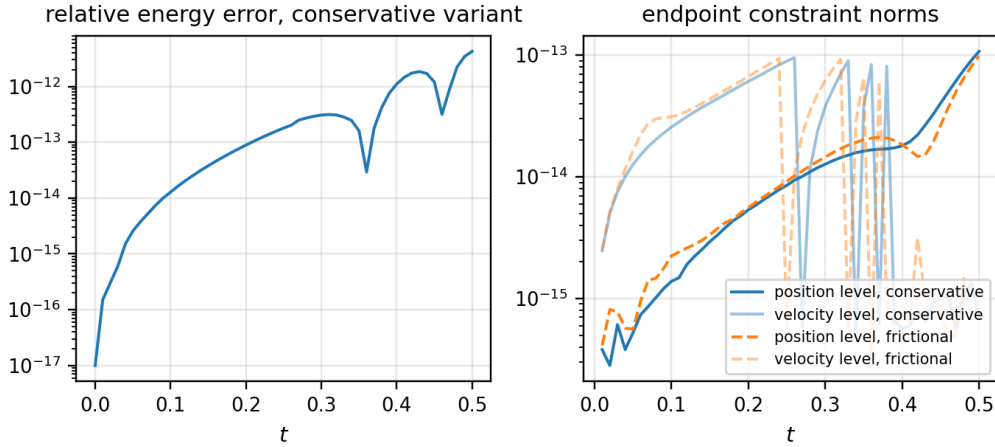


Figure 9: Chain over the regular branch $[0, 0.5]$ at $h = 0.01$: relative energy error of the conservative variant (left) and endpoint constraint norms of both variants (right; the velocity-level closure is applied only when the raw defect exceeds 10^{-13} , hence the sawtooth).

7. Limitations

1. **Scope of the identity.** Lemma 1 is stated for the open chain, where the joint coordinates form a global chart. For closed loops the same rows are solved, but the identity holds only on local charts of independent coordinates, and the theorem does not by itself turn the roundoff-level constraint residuals observed on the four-link and slider-crank

mechanisms into a trajectory-error bound; that needs a stability constant for the mechanism at hand.

2. **Regular branch.** The frozen sliding direction of the chain’s second pair is a modelling choice that is regular only while $n_1 \cdot a_1(q) \neq 0$; on the benchmark initial state this holds up to $t \approx 0.6$, and all chain experiments stop at $T = 0.5$. A pair whose sliding direction follows the axis would remove the limit at the cost of a different (still smooth) stage system; the analysis applies unchanged.
3. **Sharp friction.** The theorem’s constants and its threshold h_0 are governed by the curvature of the regularized friction law. For Stribeck velocities below about 0.1 the observed order is reduced on coarse grids and recovers only for $h \lesssim 3 \times 10^{-3}$ (Section 6). Set-valued Coulomb friction is outside the smoothness hypothesis.
4. **Solver hypothesis.** Lemma 7 covers predictors within $O(h)$ of the Gauss stage. The implemented predictor is not of that kind, and the experiments use a fixed Newton tolerance rather than the h^7 -scaled rule, so on the finest grids they satisfy (H3) only with a large constant c_η ; that the accepted iterates are nevertheless the Gauss roots to within 10^{-11} is checked a posteriori, not guaranteed a priori.
5. **Cost.** The reference implementation forms and factorizes a dense 132×132 Jacobian by automatic differentiation in Python; wall-clock times are dominated by interpreter overhead and are reported only as indications. A production implementation would exploit the stage-block structure and the sparsity of the constraint rows; no scalability study in the number of bodies is reported.
6. **Comparisons.** The time finite-element method of [1] has not been implemented; the comparison with it is formal (order six against order five). The public absolute-coordinate codes are run as published, with their own tolerances, and are compared on the double pendulum only, the one benchmark mechanism whose motion is not prescribed.

8. Conclusions

The Gauss6/FullVA step solves a single square stage system that couples three-stage Gauss collocation of the joint coordinates, the lower-pair constraints at position, velocity and acceleration level, and the Newton–Euler balance, and reconstructs the rotational endpoint through the exponential

map. The stage system is exactly reduced Gauss collocation lifted to absolute coordinates: the lifted reduced Gauss stage satisfies all 132 rows, and every root of the non-dynamic rows on the regular branch is such a lift. The sixth order of the method is therefore inherited from Gauss collocation, and the implementation-side perturbations, endpoint closure and inexact Newton, are $O(h^7)$ under a smoothness hypothesis alone; the interfaces through which the implementation enters are derived for small steps, with a threshold that is set by the friction law. The algebra and the perturbation chain are machine-checked.

The experiments confirm the picture at every point where it can be tested: sixth order in position, orientation, velocity and angular velocity over the regular branch of a frictional chain, with constraint norms at the 10^{-13} level and energy conservation of the frictionless variant to 10^{-12} ; orders two, four and six for the one-, two- and three-stage members with the same stage system at the same Newton work; sixth order on the ASME double pendulum and roundoff-level reproduction of the three driven mechanisms; and a controlled loss of order when the friction law is sharpened, in the direction the analysis predicts. Against the public absolute-coordinate codes on the double pendulum over the public horizon, the method reaches errors below 10^{-10} at step sizes where those codes still have errors of order one.

Three directions follow. The identity suggests that other stage-level constraint hierarchies (Lobatto, Radau) inherit their reduced-problem order in the same way, which would give L -stable members for stiff joints. The $O(h)$ predictor of Lemma 7 would remove the last a-posteriori element of the analysis at no cost. And a sparse, block-structured implementation would turn the per-step cost into one comparable to that of the low-order absolute-coordinate schemes, at which point the work/precision advantage measured here in Newton iterations would appear in wall-clock time as well.

CRedit authorship contribution statement

Jingquan Wang: Conceptualization, Methodology, Software, Validation, Formal analysis, Writing – original draft, Writing – review and editing.

Declaration of competing interest

The author declares no competing interests relevant to this work.

Funding

No external funding was received for this research.

Data availability

The implementation, the experiment scripts that generate every table and figure of Section 6, the result files, and the Lean 4 development are available in the repository described in Appendix D.

Declaration of generative AI and AI-assisted technologies in the writing process

During preparation of this work, the author used an AI coding assistant for drafting support, consistency checking and script editing. The author reviewed and edited the content and takes full responsibility for the published article.

Appendix A. Lower-pair constraint rows of the benchmark mechanisms

The four benchmark mechanisms use the scalar constraint rows of the public suite [32]. Let body i have centre position r_i , rotation matrix $A_i \in SO(3)$, translational velocity v_i , angular velocity ω_i , translational acceleration a_i and angular acceleration α_i . A marked point and a marked direction fixed in body i are $x_i = r_i + A_i s_i$ and $u_i = A_i a_i$. The row families are

$$C_{CD,k}(q, t) = e_k^T (x_j - x_i) - d_k(t), \quad C_{DP1}(q, t) = u_i^T u_j - c(t), \quad (\text{A.1})$$

$$C_{DP2}(q, t) = u_i^T (x_j - x_i) - d(t), \quad C_D(q, t) = \|x_j - x_i\|^2 - \ell^2(t), \quad (\text{A.2})$$

where CD is a coordinate difference, DP1 constrains two body-fixed directions, DP2 constrains a body-fixed direction against a point difference, and D is a distance row. Revolute joints are assembled from CD and DP1 rows; the slider-crank prismatic joint additionally uses DP2 and distance rows; drivers prescribe $c(t)$ of a DP1 row. With $\dot{x}_i = v_i + \omega_i \times A_i s_i$, $\dot{u}_i = \omega_i \times u_i$, $\ddot{x}_i = a_i + \alpha_i \times A_i s_i + \omega_i \times (\omega_i \times A_i s_i)$, $\ddot{u}_i = \alpha_i \times u_i + \omega_i \times (\omega_i \times u_i)$, $\Delta x = x_j - x_i$

and its derivatives, the velocity- and acceleration-level rows enforced at every stage are

$$\dot{C}_{\text{CD},k} = e_k^T \Delta \dot{x} - \dot{d}_k, \quad \ddot{C}_{\text{CD},k} = e_k^T \Delta \ddot{x} - \ddot{d}_k, \quad (\text{A.3})$$

$$\dot{C}_{\text{DP1}} = \dot{u}_i^T u_j + u_i^T \dot{u}_j - \dot{c}, \quad \ddot{C}_{\text{DP1}} = \ddot{u}_i^T u_j + 2\dot{u}_i^T \dot{u}_j + u_i^T \ddot{u}_j - \ddot{c}, \quad (\text{A.4})$$

$$\dot{C}_{\text{DP2}} = \dot{u}_i^T \Delta x + u_i^T \Delta \dot{x} - \dot{d}, \quad \ddot{C}_{\text{DP2}} = \ddot{u}_i^T \Delta x + 2\dot{u}_i^T \Delta \dot{x} + u_i^T \Delta \ddot{x} - \ddot{d}, \quad (\text{A.5})$$

$$\dot{C}_{\text{D}} = 2\Delta x^T \Delta \dot{x} - 2\ell \dot{\ell}, \quad \ddot{C}_{\text{D}} = 2\Delta \dot{x}^T \Delta \dot{x} + 2\Delta x^T \Delta \ddot{x} - 2(\ell^2 + \ell \ddot{\ell}). \quad (\text{A.6})$$

Mechanism data. All bodies are steel bars of square cross-section 0.05 m and density 7800 kg/m³ unless stated otherwise; e_x, e_y, e_z are the coordinate directions and body-fixed vectors are given in the body frame. *Single pendulum:* one bar of length 4 m (mass 78 kg, $J = \text{diag}(0.0325, 104.016, 104.016)$ kg m²), ground revolute joint with CD rows $s_1 = (-2, 0, 0)$, $s_0 = 0$ and DP1 rows (e_x, e_x) , (e_y, e_x) ; driven DP1 row $(e_y, -e_z)$ with $c(t) = \cos \theta(t)$, $\theta(t) = \pi/2 + (\pi/4) \cos 2t$. *Double pendulum:* bars of length 4 and 2 m (masses 78 and 39 kg), ground revolute joint as above, inter-link revolute joint with CD rows $s_1 = (2, 0, 0)$, $s_2 = (-1, 0, 0)$ and DP1 rows on body 2 against ground (e_x, e_x) , (e_y, e_x) ; released from the horizontal configuration $(\theta_1, \theta_2)(0) = (-\pi/2, 0)$ at rest; no friction, no drive. *Four-link loop:* rotor, link 2 and link 3 with masses 2, 1, 1 kg and inertia diagonals $(4, 2, 0)$, $(12.4, 0.01, 0)$, $(4.54, 0.01, 0)$ kg m²; revolute joint A (body 1–ground, CD $s_1 = s_0 = 0$, DP1 (e_x, e_x) , (e_y, e_x)), revolute joint D (body 3–ground, CD $s_3 = (0, 3.7, 0)$, $s_0 = (-4, -8.5, 0)$, DP1 $((0, 0, 1), e_y)$, (e_y, e_y)), spherical joint (bodies 3–2, CD $s_3 = (0, -3.7, 0)$, $s_2 = (0, -6.1, 0)$), universal joint (bodies 1–2, CD $s_1 = (-2, 0, 0)$, $s_2 = (0, 6.1, 0)$, DP1 $((0, -0.662, 0.75), e_z)$); driving DP1 row $((-1, 0, 0), e_y)$ with $c(t) = \cos(\pi t + \pi/2)$. *Slider-crank:* crank, rod and slider with masses 0.12, 0.5, 2 kg and inertia diagonals $(10^{-4}, 10^{-5}, 10^{-4})$, $(4 \times 10^{-3}, 4 \times 10^{-4}, 4 \times 10^{-3})$, $(10^{-4}, 10^{-4}, 10^{-4})$ kg m²; revolute joint A (ground–crank, CD $s_4 = (0, 0.1, 0.12)$, $s_1 = 0$, DP1 (e_y, e_x) , (e_z, e_x)), spherical joint B (rod–crank, CD $s_2 = (-0.15, 0, 0)$, $s_1 = (0, 0.08, 0)$), joint C (rod–slider, DP1 (e_y, e_z) , two DP2 rows from rod point $(0.15, 0, 0)$ to slider point 0 along e_y , e_z), translational joint D (slider–ground, DP1 $(-e_x, e_x)$, (e_y, e_x) , $(-e_x, e_y)$, DP2 rows from slider point 0 to ground point $(0.2, 0, 0)$ along $-e_x$, e_y), distance row between rod point $(0.15, 0, 0)$ and slider point $(0, 0, 1)$ with $\ell = 1$; driving DP1 row (e_y, e_y) with $c(t) = \cos(-2\pi t + \pi/2)$.

Appendix B. Diagnostics of the analysis constants

Three diagnostics of Section 5 are evaluated on the chain runs of Section 6.1. (i) Each converged stage vector, obtained with a Newton tolerance of 10^{-13} , is read in the joint-coordinate chart and the reduced collocation defects $\Delta_i[s_j]$, $\Delta_i[\dot{s}_j]$, $\Delta_i[\theta_j]$, $\Delta_i[\dot{\theta}_j]$ of Eq. (8) are evaluated together with the component of $d_j, \dot{d}_j, \ddot{d}_j$ orthogonal to n_j : all are at roundoff (Table B.9), as Lemma 1 states. (ii) The constants of Lemma 3 are measured from the implemented Jacobian at the converged stage and at the root of the $h = 0$ stage system (Table B.10): the conclusion $\|J_h^{-1}\| \leq 2\|J_0^{-1}\|$ holds with ratio at most 1.55, while the sufficient Neumann condition would need $h \leq 2 \times 10^{-5}$ in the Euclidean norm and $h \leq 8 \times 10^{-4}$ in the endpoint-linearized norm; without joint friction the threshold rises to 1.4×10^{-2} . (iii) The distances of the two predictors to Z_G (Table B.11) confirm the discussion after Lemma 7: the lifted endpoint predictor is $O(h)$ from the Gauss stage, the implemented predictor is at distance about 50 and its first Newton step expands that distance by up to 18%.

Table B.9: Exact stage identity at the converged stages of the chain (Newton tolerance 10^{-13} , window $[0, 0.08]$): maximal implemented residual, maximal reduced collocation defect and maximal component of $d_j, \dot{d}_j, \ddot{d}_j$ orthogonal to n_j .

h	steps	Newton total	max stage residual	max reduced defect	max $\perp n_j$ component
0.04	2	12	6.8×10^{-14}	9.0×10^{-15}	1.5×10^{-15}
0.02	4	24	1.5×10^{-14}	5.5×10^{-16}	1.9×10^{-15}
0.01	8	48	1.6×10^{-14}	5.8×10^{-16}	2.0×10^{-15}

Table B.10: Constants of Lemma 3 on the chain: $M_0 = \max_n \|J_0^{-1}\|$, $C_J = \max_n \|J_h - J_0\|/h$, observed inverse ratio $\|J_h^{-1}\|/\|J_0^{-1}\|$ and Neumann threshold $h_0 = 1/(2M_0C_J)$, in the Euclidean norm on the stage layout (E) and in the endpoint-linearized norm $\|J_0^{-1} \cdot \|$ (L, where $M_0 = 1$); the last two columns repeat the L-norm quantities for the same mechanism without joint friction.

h	M_0 (E)	C_J (E)	ratio	h_0 (E)	C_J (L)	h_0 (L)	C_J (L, no fr.)	h_0 (L, no fr.)
0.04	18.2	7.5e2	1.55	3.7e-5	5.1e2	9.8e-4	34.1	1.5e-2
0.02	24.3	8.2e2	1.26	2.5e-5	5.6e2	8.9e-4	35.5	1.4e-2
0.01	28.4	9.6e2	1.18	1.8e-5	6.5e2	7.6e-4	36.5	1.4e-2

Table B.11: Predictor distances for Lemma 7 on the chain (Euclidean norm on the stage layout): distance of the lifted endpoint predictor $Z_*^{(0)}$ and of the implemented predictor to Z_G , and the ratio $\|Z^{(1)} - Z_G\|/\|Z^{(0)} - Z_G\|$ of the first full Newton step from the implemented predictor.

h	lifted predictor	implemented predictor	first-step ratio
0.04	3.23	50.2	1.12
0.02	2.11	50.3	1.16
0.01	1.07	50.4	1.18

Solver envelope. Table B.12 records, for every grid of the sweep, the largest residual of the accepted Newton iterate under the production stopping rule and its ratio to h^7 , which is the constant c_η with which that grid satisfies (H3). The accepted residual is set by the tolerance, not by h , so the ratio grows from 10^{-4} to 10^6 across the sweep; as discussed after Lemma 7, the resulting bound $2M\|F_h(\tilde{Z})\|$ on the stage perturbation is pessimistic by three orders of magnitude relative to the observed errors. The endpoint position-level constraint norm of Table 4 is the quantity that actually dominates the deviation of the numerical trajectory from the constraint manifold; it decreases as h^6 as the theorem requires.

Table B.12: Solver envelope on the chain sweep ($\tau_N = 10^{-11}$): largest residual norm of the accepted iterate over all steps, its ratio to h^7 (the c_η of (H3) for that grid), and the largest components of the stage displacements $d_j, \tilde{d}_j, \ddot{d}_j$ orthogonal to n_j .

h	Newton/step	$\max_n \ F_h(\tilde{Z}_n)\ $	ratio to h^7 (c_η)	$\max \perp n_j$ component
0.1	6.00	8.1e-12	8.1e-5	5.6e-13
0.05	6.00	3.5e-13	4.5e-4	3.2e-15
0.025	5.90	8.8e-12	1.4	6.4e-14
0.0125	5.72	2.7e-12	5.7e1	4.3e-15
0.00625	5.72	3.1e-12	8.3e3	4.1e-15
0.003125	5.72	3.4e-12	1.2e6	5.3e-15

Appendix C. Lean development

The Lean 4 development (Mathlib pinned by its manifest) machine-checks the algebraic and perturbation lemmas of Section 5: 57 theorems, no open

goals, and an exhaustive axiom audit showing that every theorem depends only on the standard axioms `propext`, `Classical.choice` and `Quot.sound`. It builds with `lake build`; the script `scripts/check.sh` runs the build, the axiom audit and a lint pass. Table C.13 maps the statements of the paper to the theorems that check them. The row identities are proved for a transcription of the implemented residual into Lean; the perturbation lemmas are proved in the abstract normed-space setting with the constants of the paper. Butcher’s order theorem and the Gauss local-error estimate are not formalized and enter as cited results.

Table C.13: Statements of the paper and the Lean theorems that check them.

Statement	Lean theorem
Lemma 1, non-dynamic rows (both directions)	<code>FullVA.Transition.nondynamic_rows_iff</code>
Lemma 1, Newton–Euler rows	<code>NewtonEuler.dynamic_rows_vanish</code>
Lemma 3, bound Eq. (18)	<code>uniform_inverse_of_perturbation</code>
Lemma 2, root uniqueness	<code>root_unique_of_linearization</code>
Lemma 4, bound Eq. (20)	<code>gauss6_step_defect</code>
Butcher conditions $B(6)$, $C(3)$, $D(3)$; failure of $B(7)$	<code>Gauss6.butcher_order_six_hypotheses</code> , <code>Gauss6.not_B_seven</code>
Lemma 5	<code>endpoint_closure_exists</code> , <code>endpoint_correction_bound_h7</code>
Lemma 6	<code>inexact_newton_output_bound</code>
Lemma 7, decay Eq. (23)	<code>simplified_newton_residual_decay</code>
Lemma 8	<code>local_to_global</code> , <code>reported_grid_bound</code>
Theorem 1, three-term chain and grid bounds	<code>local_defect_bound</code> , <code>conditional_sixth_order_grid_bound</code>

Appendix D. Reproducibility

The repository accompanying the paper has five top-level folders: `paper/` (this manuscript), `proof/` (the Lean development), `numerics/` (the implementation and all experiment scripts), `external/` (the public benchmark codes as git submodules or mirrors) and `validation/` (the audit trail of the development, including the check that the numbers in this paper equal the values in the result files). The implementation of the step is the single residual function in `numerics/v047_cylindrical_chain_pipeline/run_v047.py`; the experiments of Section 6 are the scripts `e1_convergence.py` to `e7_solver_envelope.py` in `numerics/v049_paper_experiments/`,

each of which writes a CSV and a JSON file under `results/`; the figures are drawn from those files by `make_figures.py`, and `validation/validate_manuscript.py` checks that every number in the tables of that section equals the value in its result file. The Lean development is checked by `proof/scripts/check.sh`. All runs reported were made on one CPU core; the double pendulum reference at $h = 10^{-4}$ over $T = 3$ took 1.6 h, each of the other scripts between a minute and about an hour.

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